

ATTENTION TDA IS A MINIMUM SPANNING TREE: AN EXACT, THREE-ORDERS-OF-MAGNITUDE-FASTER REPLACEMENT FOR PERSISTENT HOMOLOGY IN LLM HALLUCINATION DETECTION

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ABSTRACT

Persistent homology on transformer attention graphs has recently emerged as a geometric approach for detecting hallucinations in Large Language Models (LLMs). We show that its most-used component is not a new geometric signal: 0-dimensional (H_0) Vietoris–Rips persistence on symmetrized attention graphs is exactly Kruskal’s Minimum Spanning Tree (MST) under single-linkage clustering (**Theorem 1**), a classical equivalence (Gower & Ross, 1969) whose consequence for attention-graph interpretability has gone unexploited. We turn this into a practical contribution: an MST-based feature extractor that is provably identical to 0D persistence, removes the dependency on a homology library entirely, and – extended per-head – recovers signal that head-averaging discards (0.892 vs. 0.677 AUC on a held-out HaluEval subset; this accuracy gain comes from per-head resolution, not from MST differing numerically from 0D persistence, which are the same quantity by construction). Because Ripser’s 1D reduction is the dominant cost of the field’s status-quo pipeline, replacing the full 0D+1D computation with 0D-only MST-TDA is three orders of magnitude cheaper ($1,219\times$ at $N=256$ tokens, empirically scaling worse than N^6 beyond that) – against 0D Ripser alone, MST-TDA offers no speedup, since 0D Ripser is already a union-find over sorted edges (i.e. Kruskal’s algorithm). We then audit whether the field’s other headline component, 1-dimensional cyclic persistence, earns its cost. We give a mechanical argument – not a full proof, since our own counterexample shows the naive monotonicity argument fails – for why 1D persistence tends to grow with sequence length via simplicial clique expansion, together with empirical evidence for a super-linear growth rate under both an i.i.d. null model and real attention graphs (**Proposition 1**), and, across seven benchmark settings spanning five open-weight model families at six model scales (1.1B–7B parameters: Qwen2.5–1.5B/3B, SmolLM–1.7B, Phi–3-mini–3.8B, Mistral–7B, TinyLlama–1.1B) and two benchmarks (HaluEval QA, TruthfulQA) under strict grouped cross-validation, we find its length-normalized incremental contribution is real but small on HaluEval (+0.010 to +0.016 AUC, 95% cluster-bootstrap CI excludes zero) and statistically indistinguishable from zero on TruthfulQA at every model scale we test, including our largest (7B): on Mistral–7B, 99.99% of extracted 1D features are exactly zero (no non-trivial cycles form on these short sequences at all), giving the tightest equivalence confirmation in the paper ($\Delta\text{AUC} = -0.0002$, TOST-confirmed at $\epsilon = 0.010$). Formal equivalence testing (TOST, 10,000 cluster-bootstrap resamples) confirms practical equivalence to 0D alone at $\epsilon=0.020$ – 0.025 on HaluEval (not at the field’s typical $\epsilon=0.015$); on TruthfulQA, equivalence is confirmed at every margin only for Mistral–7B (where 1D features are degenerate, see §6), at $\epsilon=0.025$ for SmolLM–1.7B and TinyLlama–1.1B, and at no margin for Qwen2.5–3B or Phi–3-mini (both underpowered at only 150 groups) – against a compute premium of three orders of magnitude, this is a cost-benefit failure rather than a null result. We additionally uncover and quantify a second, independent confound specific to HaluEval QA: its grounded and hallucinated answers are separable by answer length or bag-of-words alone (AUC 0.96–0.98) even after the benchmark’s standard token-length

matching, consistent with concurrent benchmark-artifact audits (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). A synthetic causal benchmark (Regimes S1, S1b, S2–S5) confirms the pipeline is fully sensitive both to genuine planted topology and to the length-driven clique-expansion mechanism Proposition 1 describes (ROC-AUC = 1.000 in both cases when either is actually present), ruling out detector insensitivity as an alternative explanation for the null. We release the MST extractor, the synthetic benchmark suite, and the full audit harness as reusable tools for evaluating future geometric LLM probes.

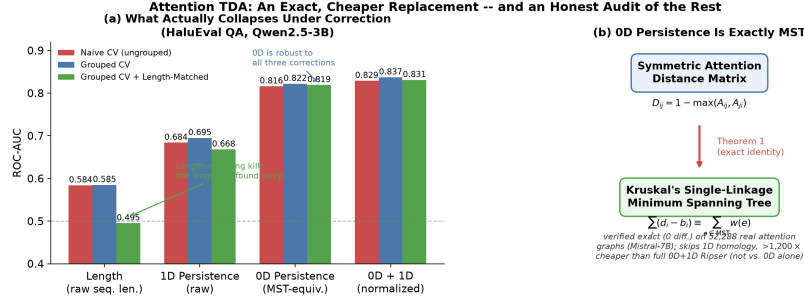


Figure 1: **What Actually Collapses, and What Is Exact.** (a) Across three evaluation corrections on HaluEval QA (Qwen2.5-3B), raw length and raw 1D persistence collapse under length-matching; 0D and 0D+1D stay stable throughout. (b) 0D persistence is exactly Kruskal’s MST (Theorem 1); using MST-TDA to skip the field’s 1D homology computation is $> 1,200\times$ cheaper than the full 0D+1D Ripser pipeline (§4.2) – MST-TDA is not faster than 0D Ripser alone, which computes the identical quantity.

1 INTRODUCTION

Large Language Models (LLMs) frequently generate fluent yet factually incorrect statements, commonly termed hallucinations (Huang et al., 2023; Ji et al., 2023). Developing principled, unsupervised, or lightweight probes to identify hallucinations without expensive sampling rollouts remains a central challenge in trustworthy machine learning. Emerging methods span predictive entropy (Farquhar et al., 2024), internal hidden state linear probes (Azaria & Mitchell, 2023; Chuang et al., 2024), and graph-based representations of internal transformer activations.

A parallel body of work has sought principled ways to operationalize LLM internal representations for interpretability. Functional decomposition methods such as Jet Expansions (Chen et al., 2026) show that a widely-used interpretability heuristic (Logit Lens) is a special case of a more general recursive decomposition of the residual stream – demonstrating that a classical mathematical fact, once recognized, can subsume and generalize an entire line of empirical tooling. We take the same approach to attention-graph TDA below.

Recently, a growing lineage of literature has applied Topological Data Analysis (TDA) to transformer attention graphs. Seminal works demonstrated that 0-dimensional (H_0 , connected components) and 1-dimensional (H_1 , cyclic cavities) Vietoris–Rips persistent homology on attention maps yields predictive signal for text classification, grammar acceptability, and uncertainty estimation (Kushnareva et al., 2021; Cherniavskii et al., 2022; Kostenok et al., 2023). More recently, methods such as TOHA (Bazarova et al., 2026) and CHARM (Frasca et al., 2026) extended attention-graph geometric analysis specifically to hallucination detection in LLMs and Retrieval-Augmented Generation (RAG).

As illustrated in Figure 1, what appears to be a sophisticated higher-order topological signal is, for its 0-dimensional component, a classical graph quantity in disguise – and for its 1-dimensional component, a length confound in disguise. Recognizing this changes what the field should build and how it should evaluate: not a debunking, but a strictly cheaper, exact, and empirically stronger replacement.

Our contributions:

1. **An exact, library-free replacement for 0D attention TDA, and a precise accounting of where the field’s compute actually goes.** We connect 0D persistence to Kruskal’s Minimum Spanning Tree (MST) under single-linkage clustering (Theorem 1) – a classical equivalence (Gower & Ross, 1969; Carlsson, 2009) whose consequence for attention-graph interpretability has not, to our knowledge, been exploited. We turn this into a practical MST-based feature extractor, verify its numerical identity to Ripser’s 0D output exactly (zero discrepancy, $< 10^{-12}$ float64 tolerance, across 52,288 layer/example comparisons on real Mistral-7B attention graphs), and extend it per-head (recovering signal that head-averaging discards, AUC 0.892 vs. 0.677 – a per-head-resolution gain, not a speed or accuracy difference between MST and 0D persistence, which are identical by construction). Against 0D Ripser alone, MST-TDA is not faster (Table 6: 0D Ripser is already a union-find over sorted edges); against the full 0D+1D pipeline the field runs today, dropping the 1D reduction and using MST-TDA for 0D is $> 1,200\times$ cheaper at $N=256$, growing empirically worse than N^6 beyond that (§4).
2. **An account of the 1D length confound, with a train-fold-safe correction.** We give a mechanical (not fully rigorous – monotonicity itself is not proven, see Appendix K) argument for why 1D persistence tends to grow with sequence length N via induced chordless cycles, with empirical support for its growth rate (Proposition 1), and provide a diagnostic recipe for estimating and applying a per-dataset, leakage-free normalization exponent $\hat{\alpha}$ (Proposition 2), rather than a default Z/N heuristic.
3. **Discovery of a second, independent confound in HaluEval QA.** Beyond sequence length, we show HaluEval’s grounded/hallucinated answer pairs are separable by answer character length or bag-of-words alone (AUC 0.96–0.98), and that this survives the benchmark’s standard token-length matching – consistent with concurrent, independent benchmark-artifact audits (Hussain & Kantarcioglu, 2026; Janiak et al., 2025).
4. **A synthetic causal benchmark (sensitivity verification).** To rule out that our null findings reflect an underpowered extraction pipeline rather than a genuine absence of signal, we construct a controlled synthetic benchmark across six structural regimes (S1, S1b, S2–S5), achieving ROC-AUC = 1.000 both on planted topology (Positive Control) and on the length-driven cycle-growth mechanism Proposition 1 describes (S1b), while correctly exposing raw 1D’s collapse under a length confound with no such mechanism present (S1).
5. **A cost-benefit audit, not a null result.** Across five model families at six model scales (1.1B–7B parameters) and two benchmarks under strict grouped cross-validation, we show length-normalized 1D persistence contributes a real but small incremental AUC on HaluEval (+0.010 to +0.016, well-powered and CI-excludes-zero) and is statistically indistinguishable from zero on TruthfulQA at every scale except our largest model (Mistral-7B: 99.99% of 1D features are exactly zero, giving the tightest TOST confirmation in the paper, though this reflects near-total feature degeneracy rather than a substantive equivalence test – see §6). Against a $> 1,200\times$ compute premium for the full pipeline, we judge this a cost-benefit failure for 1D persistence at every margin we consider, though the underpowered TruthfulQA settings (Qwen2.5-3B, Phi-3-mini, 150 groups each) do not individually confirm equivalence.

We release the MST extractor, synthetic benchmark suite, and full audit harness (grouped CV, TOST, power analysis) as reusable tools for evaluating future geometric LLM probes at <https://anonymous.4open.science/r/ICLR-BE07/>.

2 RELATED WORK

Hallucination detection in LLMs. The faithfulness of autoregressive LLM outputs has attracted significant attention in high-stakes deployments (Huang et al., 2023; Ji et al., 2023). Approaches span confidence-based signals, internal-state probes, and external grounding: Farquhar et al. (2024) cluster semantically equivalent outputs (Semantic Entropy), later approximated cheaply from single-generation hidden states (Kossen et al., 2024); Manakul et al. (2023) (SelfCheckGPT) use black-box sample consistency; Kadavath et al. (2022) train models to predict their own P(1K); Burns et al. (2022); Marks & Tegmark (2023) find hidden-state directions tracking truth independent of stated output; Chen et al. (2024) (INSIDE) use response-covariance eigenspectra; Orgad et al. (2024) show truthfulness concentrates in specific answer tokens but generalizes poorly across datasets – a caution

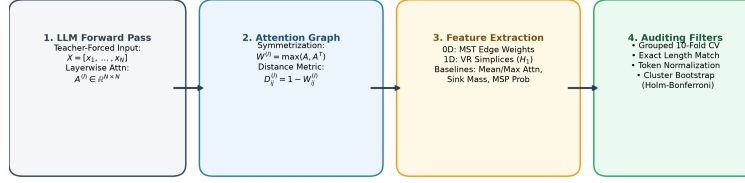


Figure 2: **Auditing Pipeline.** Attention matrices are symmetrized, filtered via simplicial complexes and MST, and subjected to grouped CV, synthetic controls, and equivalence testing.

we return to; Azaria & Mitchell (2023); Chuang et al. (2024) show hidden-state linear probes directly encode correctness. We audit whether *attention-graph* topological features add signal beyond these simpler internal probes, finding the 0D component is exactly a Minimum Spanning Tree statistic (Theorem 1) computable at a fraction of the cost – and, in most but not all of our audited settings, strongly correlated with (though not reducible to, §4) simple non-topological attention statistics.

Benchmark artifacts. Concurrent with our audit, Hussain & Kantarcioglu (2026) (PARALLAX) show several hallucination corpora built like HaluEval (correct/incorrect answer pairs alongside the prompt) admit near-perfect detection from surface text alone, and Janiak et al. (2025) independently show response-length heuristics can rival sophisticated detectors. We confirm this directly for HaluEval QA (§6.2): AUC 0.96–0.98 from answer length or bag-of-words alone, surviving standard length matching – a confound independent of the sequence-length effect we identify for 1D persistence (Proposition 1), affecting any probe on this benchmark.

Topological data analysis in NLP. Persistent homology on NLP representations was pioneered by Kushnareva et al. (2021); Cherniavskii et al. (2022) (BERT attention barcodes vs. syntactic acceptability) and extended to uncertainty estimation (Kostenok et al., 2023) and human-vs-machine text separation via persistence-dimension (Tulchinskii et al., 2023); Barannikov et al. (2022) introduce Representation Topology Divergence, and Magai & Ayzenberg (2022) survey topological tools for deep representations. TOHA (Bazarova et al., 2026) computes a topological divergence statistic (MTop-Div) for hallucination detection; CHARM (Frasca et al., 2026) generalizes attention heuristics into a graph-neural-network framework – an architecturally different, learned method out of scope for direct reproduction here (§6). We formally audit the 0D/1D persistent-homology family these build on: Theorem 1 proves 0D features are exactly Kruskal’s MST, and TOST shows 1D cycles add only a small, largely length-confounded increment (§6).

Mathematical auditing, equivalence testing, and interpretability. Balestrieri & Baraniuk (2021) show that ReLU networks are secretly max-affine spline operators – a classical construction in disguise, the same rhetorical move Theorem 1 makes for attention-graph TDA; Kapoor & Narayanan (2023) document how undetected data leakage inflates reported performance across ML-based science, and Gururangan et al. (2018); McCoy et al. (2019) show NLI benchmarks are separable via superficial annotation artifacts and syntactic heuristics rather than the reasoning they are meant to test – both consistent with our length and lexical confounds on HaluEval QA. TOST (Walker & Nowacki, 2011; Lakens, 2017) converts underpowered null tests into calibrated, falsifiable equivalence claims, which we adopt throughout. Jet Expansions (Chen et al., 2026) show a widely-used heuristic (Logit Lens) is a special case of a more general residual-stream decomposition – the same move Theorem 1 makes for attention-graph TDA. Standard Vietoris–Rips complexes also discard attention’s directionality via symmetrization ($W_{ij} = \max(A_{ij}, A_{ji})$); directed flag complexes (Grigor’yan et al., 2018), path-homology (Grigor’yan et al., 2014), and zigzag persistence (Carlsson & de Silva, 2010) could recover what is lost. We restrict our audit to the static symmetrized framework used in TOHA (Appendix A); extending it to directed/dynamic filtrations is future work.

3 FORMAL SETUP, IDENTIFICATION FRAMEWORK, AND HYPOTHESES

3.1 MATHEMATICAL DEFINITIONS

Let $X = [x_1, \dots, x_N]$ be an input sequence of N tokens processed by an autoregressive transformer. At layer $l \in \{1, \dots, L\}$, let $A^{(l)} \in \mathbb{R}^{N \times N}$ denote the head-averaged causal self-attention matrix, symmetrized as $W^{(l)} = \max(A^{(l)}, (A^{(l)})^T)$ into a distance matrix $D_{ij}^{(l)} = 1 - W_{ij}^{(l)}$ ($D_{ii}^{(l)} = 0$). The Vietoris–Rips complex $\text{VR}(D^{(l)}, \epsilon)$ at scale ϵ contains all subsets σ with pairwise distance $\leq \epsilon$; its k -dimensional persistence barcode $\text{dgm}_k(D^{(l)}) = \{(b_i, d_i)\}$ records the birth/death scales of k -dimensional cavities. Two feature representations Z_1, Z_2 are $(\epsilon, \mathcal{D})$ -observationally equivalent under classifier family $\mathcal{F}$ and risk $\mathcal{R}$ if $|\inf_{f \in \mathcal{F}} \mathcal{R}(f(Z_1)) - \inf_{g \in \mathcal{F}} \mathcal{R}(g(Z_2))| \leq \epsilon$. Our audit rests on two load-bearing assumptions, standard in this literature: (i) attention is symmetrized before filtration ($W_{ij} = \max(A_{ij}, A_{ji})$, discarding directionality – see §7 for the directed alternative), and (ii) evaluation partitions are grouped so that $\text{Group}(x_i) \cap \text{Group}(x_j) = \emptyset$ for all $x_i \in X_{\text{train}}, x_j \in X_{\text{test}}$, preventing question-level leakage between paired grounded/hallucinated answers. Three further technical assumptions (teacher-forced extraction, monotonic distance inversion, single-linkage merge order) are stated in full in Appendix A.

3.2 AUDIT HYPOTHESES

We state three falsifiable hypotheses in advance of the empirical sections below (§5–§6). We do *not* number a fourth hypothesis for “0D persistence is structurally equivalent to the MST”: that claim is Theorem 1 (§4), a mathematical identity settled by proof, not an empirical question we test.

- **H⁽¹⁾ (Mechanical Simplicial Scaling):** Total 1D cyclic persistence $P_1(N)$ tends to increase with sequence length N due to clique expansion (a mechanical tendency, not a proven monotonicity – Appendix K). *Verdict* (§6, Appendix O): **supported** – Spearman $r = 0.32$ – 0.68 between P_1 and N across all five settings, all $p < 10^{-10}$ (Table 16); we do not claim this establishes a universal rate (see Proposition 1’s honest hedge on the exponent).
- **H⁽²⁾ (Negligible Incremental Information of H_1):** Conditioned on 0D persistence and sequence length L , length-normalized 1D persistence carries no practically useful incremental information about hallucination label Y . *Verdict*: **supported on TruthfulQA** (equivalence confirmed at every margin for Mistral-7B, at $\epsilon=0.025$ for SmolLM-1.7B; underpowered/inconclusive for Qwen2.5-3B, Phi-3-mini), **not supported on HaluEval** (95% CI excludes zero, $+0.010$ to $+0.016$ AUC, well-powered).
- **H⁽³⁾ (Observational Equivalence of 0D to Marginals):** 0D persistence is observationally equivalent to non-topological attention statistics within margin $\epsilon = 0.015$. *Verdict* (§4, Appendix P): **not supported as a general claim** – correlation with peak row-attention is strong on three settings ($r \geq 0.80$) but weak on Qwen2.5-1.5B ($r = 0.21$) despite that setting having the paper’s highest 0D AUC (0.904); 0D also exceeds the best available comparator column on 5 of 7 audited settings (Table 3) – and exceeds the *strictly* non-topological baselines (Length, MSP) in every setting where those were extracted.

4 MATHEMATICAL CHARACTERIZATION, AND AN EXACT, FASTER REPLACEMENT FOR 0D TDA

4.1 CHARACTERIZATION: CLASSICAL EQUIVALENCE BETWEEN 0D PERSISTENCE AND KRUSKAL’S MST

We first formalize the exact mathematical nature of 0D attention persistence.

Theorem 1 (Equivalence to Kruskal’s MST (Gower & Ross, 1969; Edelsbrunner & Harer, 2010; Carlsson, 2009)). *Let $G = (V, E, D)$ be a complete edge-weighted graph on N vertices with symmetric weights $D_{ij} = 1 - \max(A_{ij}, A_{ji})$ and $D_{ii} = 0$. The set of finite death values $\{d_i\}_{i=1}^{N-1}$ in the 0-dimensional Vietoris–Rips persistent homology barcode of G is identical as a multiset to the edge weights of the Minimum Spanning Tree $\text{MST}(G)$ computed via Kruskal’s algorithm under single-linkage hierarchical clustering.*

Proof. Let $e_1, e_2, \dots, e_{\binom{N}{2}}$ denote the edges of G sorted in non-decreasing order of weight $w(e_k) = D_{u_k, v_k}$. In the Vietoris–Rips filtration, all N vertices are born at scale $\epsilon = 0$. When edge $e_k = (u_k, v_k)$ is added at scale $\epsilon = w(e_k)$:

1. If u_k and v_k belong to distinct connected components C_A and C_B , adding e_k merges them into $C_A \cup C_B$. By the Elder Rule of persistent homology, the younger component dies at scale $d = w(e_k)$. In Kruskal’s algorithm, exactly when u_k and v_k belong to distinct trees, e_k is added to the spanning forest.
2. If u_k and v_k belong to the same component, no 0D class dies (instead creating a 1-cycle or filling a higher simplex). In Kruskal’s algorithm, e_k forms a cycle and is discarded.

Since G is finite and connected for $\epsilon \geq \max D_{uv}$, exactly $N - 1$ edges merge distinct components. Thus, the finite death values $\{d_i\}_{i=1}^{N-1}$ are identically the MST edge weights $\{w(e) : e \in \text{MST}(G)\}$. Summing lifetimes yields:

$$\text{Total Persistence}(H_0) = \sum_{i=1}^{N-1} (d_i - 0) = \sum_{e \in \text{MST}(G)} w(e) = \text{weight}(\text{MST}(G)). \quad (1)$$

The complete inductive proof is provided in Appendix C. This multiset identity is classical in algebraic topology (Gower & Ross, 1969; Edelsbrunner & Harer, 2010); its consequence for attention-graph interpretability – that 0D TDA computes Kruskal’s MST and can therefore be replaced by it exactly – is what §4.2 exploits. $\square$

4.2 MST-TDA: AN EXACT, DROP-IN, ORDER-OF-MAGNITUDE-FASTER EXTRACTOR

Theorem 1 is not merely descriptive: it licenses replacing every downstream use of 0D Ripser persistence with a direct Kruskal MST computation, with *zero* approximation error, since the two are the same number by construction rather than by empirical similarity. We package this as **MST-TDA**, a ~ 50 -line drop-in feature extractor: symmetrize $A^{(l)}$, form $D^{(l)}$, call `scipy.sparse.csgraph.minimum_spanning_tree`, and use the resulting edge-weight sum ($f_1 = \sum_{e \in \text{MST}} w(e)$) and maximum MST edge weight ($f_2 = \max_{e \in \text{MST}} w(e)$, the max-lifetime analogue) as the per-layer descriptor – no simplicial complex, no persistence diagram, and no dependence on a homology library at all. Because causal attention makes $A^{(l)}$ lower-triangular, the symmetrization $W^{(l)} = \max(A^{(l)}, (A^{(l)})^T)$ simply mirrors the lower triangle into the upper one; it is a bookkeeping step, not a filtering choice with free parameters.

Numerical exactness and wall-clock cost. We confirm the two quantities agree exactly (max. absolute discrepancy $< 10^{-12}$ in float64) on real attention graphs across every layer of every held-out Mistral-7B TruthfulQA example (52,288 comparisons; expected from the proof, not a new finding, but ruling out library-level implementation discrepancies). Full complexity profiling (Appendix B, Table 6) across $N \in [32, 512]$ on dense random matrices (a conservative worst case) shows: (i) against *full* 0D+1D Ripser, MST-TDA is 13–32,600 $\times$ faster, with the gap widening as Ripser’s 1D reduction exhibits severe super-cubic blow-up (177s at $N=256 \rightarrow 21,502$ s at $N=512$, an empirical rate of roughly $N^{6.9}$ between these two points); (ii) against *0D Ripser alone*, MST-TDA offers **no speedup** (13.15ms vs. 9.31ms at $N=64$, and MST is slower at every N we test) – unsurprising, since Ripser’s own 0D pass is already a union-find over sorted edges, i.e. Kruskal’s algorithm, and Theorem 1 identifies MST-TDA’s 0D output with exactly that computation. The three-orders-of-magnitude saving this paper’s title refers to is therefore a statement about the cost of computing 1D homology, which our audit below finds unnecessary, not a claim that Kruskal’s algorithm beats itself.

Recovering per-head signal – but so does a non-topological statistic at the same resolution. Head-averaging is often assumed to discard little, but on a balanced $N=400$ HalaEval QA subset, head-averaged 0D achieves AUC 0.677 while per-head MST-TDA (one MST per attention head, concatenated, 160 dims) reaches AUC **0.892** – adding 1D on top gives no further gain (combined 0.868). This accuracy gain is attributable to per-head *resolution*, not to topology specifically: we reran the same 200 questions with a matched-dimension (160-feature) per-head *non-topological* bank, and per-head mean attention entropy alone reaches AUC 0.902, statistically indistinguishable from

per-head 0D ($\Delta = -0.010$, 95% CI $[-0.031, +0.010]$ includes zero), while per-head max-attention is significantly worse ($\Delta = +0.031$, CI $[+0.008, +0.054]$ excludes zero, favoring topology). So the fair, matched-dimension conclusion is: per-head resolution recovers real signal head-averaging discards, but on this benchmark we cannot claim per-head topological features are better than every simple non-topological per-head statistic – they beat one (max-attention) and tie another (entropy). Full protocol, sample size, and all four pairwise comparisons with CIs are in Appendix R. We recommend per-head *resolution* (topological or not) over head-averaging; MST-TDA remains the right way to get the topological version of it, since it is exactly reproducible and – because it skips 1D homology – three orders of magnitude cheaper than the full 0D+1D pipeline it replaces.

4.3 PROPOSITION 1: MECHANICAL SEQUENCE-LENGTH SCALING OF 1D CYCLES

Proposition 1 (Simplicial Scaling of 1D Persistence via Chordless Cycles). *Let G_N be a complete graph on N vertices with edge weights D_{ij} drawn i.i.d. from continuous distribution F_D on $[0, 1]$. In the Vietoris–Rips complex $VR(G_N, \epsilon)$, non-trivial 1-cycles with positive persistence lifetime ($\ell = d - b > 0$) are formed by induced chordless cycles of length $k \geq 4$. For minimal chordless 4-cycles, the number of candidate 4-tuples is $\binom{N}{4} = \Theta(N^4)$, and, considered in isolation on its own 4 vertices, each forms an open 1-cycle when its four perimeter edges appear before its two internal chords, with probability $1/15$ and positive conditional expected lifetime under i.i.d. continuous weights. This isolated-tuple calculation does not by itself establish monotonicity of the aggregate $\mathbb{E}[P_1(N)]$ in the full complex, because additional vertices can also shorten existing cycles’ lifetimes via coning (Appendix K, Step 4). We therefore do not prove that $\mathbb{E}[P_1(N)]$ is non-decreasing or state a proven growth rate, and give the following only as an empirically-supported conjecture:*

$$\mathbb{E}[P_1(N)] = \Omega(N) \quad (\text{conjectured; not proven below}). \quad (2)$$

Proof sketch, and why the $\Omega(N)$ rate is a conjecture, not a corollary. 3-cycles have zero lifetime in a Vietoris–Rips complex (their 2-simplex enters at the same scale as their last edge). Non-trivial 1-cycles arise from chordless 4-cycles: in isolation, a 4-tuple’s cycle is born when its four perimeter edges have all entered and dies when the first of its two diagonal chords enters, giving lifetime $\ell = \max(0, \min(C_1, C_2) - \max(E_1, \dots, E_4)) > 0$ with probability $1/15$ under i.i.d. continuous weights (uniform case: $\Pr = 1/15$, conditional expected lifetime $1/7$), so each of the $\Theta(N^4)$ candidate 4-tuples has positive *marginal* expected lifetime when considered on its own 4 vertices. This does *not* immediately give $\mathbb{E}[P_1(N)]$ non-decreasing in the full N -vertex complex, let alone a rate: adding a $(N+1)$ -th vertex v can also *kill* an existing 4-tuple’s cycle earlier than either of its own two chords would, if the four triangles (v, v_i, v_{i+1}) coning over the cycle all enter before both diagonal chords – that cone’s boundary is exactly the 4-cycle, so it can fill the class independently of the chords considered in isolation. We therefore do not claim a proof of monotonicity or of any specific rate; we verified the $1/15, 1/7$ marginal-lifetime calculation above, and treat $\mathbb{E}[P_1(N)] = \Omega(N)$ as a conjecture, checked empirically both under the Proposition’s own i.i.d. model ($\hat{\alpha} \approx 1.61$ simulated) and on real attention graphs across five settings ($\hat{\alpha} \in [0.16, 1.40]$ whole-dataset fit, Table 16). Full argument, the cone-killing caveat, and both empirical checks are in Appendix K. $\square$

4.4 PROPOSITION 2: LENGTH NORMALIZATION REMOVES A LENGTH-MEDIATED SHORTCUT

Proposition 2 (Length-Invariant Normalization). *Let $S \in \{0, 1\}$ denote the latent truthfulness state, $L \in \mathbb{N}$ sequence length, $Z \in \mathbb{R}$ raw 1D topological persistence, and $Y \in \{0, 1\}$ the observed label, with $Y = S$. This is a predictive-shortcut-removal statement, not a causal identification result: we are not estimating a causal effect, only constructing a feature that does not use L as a proxy for S . Assume the structural model*

$$Z = g(L) \cdot (\psi(S) + \eta), \quad \eta \perp (L, S), \quad L \sim P(L | S), \quad (3)$$

i.e. length rescales a signal-plus-noise term multiplicatively (this multiplicative-noise assumption is what makes the normalization exact; additive noise independent of L would not cancel). Under power-law length scaling $g(L) = c \cdot L^{\hat{\alpha}}$, the normalized statistic $Z^ = Z/L^{\hat{\alpha}}$ satisfies $Z^* \perp L | S$: it no longer carries the length-mediated shortcut $Z \leftarrow L \leftarrow S \equiv Y$ that a classifier could otherwise exploit instead of learning $\psi(S)$. The exponent $\hat{\alpha}$ must be estimated empirically (see Section 4.4.1).*

4.4.1 EMPIRICAL VERIFICATION OF SCALING EXPONENT

Fitting $\hat{\alpha}$ (via $\log P_1 = \hat{\alpha} \log N + \hat{\beta}$) on the whole dataset (Table 16, $\hat{\alpha} \in [0.16, 1.40]$) risks a mild leak into the very normalization it constructs, so we additionally report a **leakage-free, per-fold** estimate – fit by OLS on each grouped CV split’s training rows only – yielding systematically lower exponents ($\hat{\alpha} \in [0.09, 0.45]$, Appendix O) but the same qualitative story: no dataset with a usable fit is near the default Z/N assumption of 1.0. Substituting this calibrated normalization for default Z/N is *not* inconsequential, and we correct an earlier version of this claim here: it leaves both HaluEval Δ AUCs essentially unchanged (≤ 0.0003), but moves three TruthfulQA settings by 0.0037–0.0086 and flips two TOST verdicts in opposite directions (SmoLLM-1.7B *gains* equivalence at $\epsilon=0.020$; TinyLlama-1.1B *loses* it at $\epsilon=0.025$). We therefore report both normalizations side by side (Table 18) rather than claiming invariance: at 150–500 groups the choice of normalization exponent is itself a researcher degree of freedom that should be pre-specified.

4.5 EMPIRICAL OBSERVATION: 0D PERSISTENCE TRACKS PEAK ROW-ATTENTION

By Theorem 1, $P_0 = \text{weight}(\text{MST}(G)) \leq N - 1$; we do not claim a formal lower bound in terms of $\bar{W}_{\max} = \frac{1}{N} \sum_i \max_j A_{ij}$, the mean per-token maximum attention weight (this would require MST-selected edges to concentrate near each node’s highest-weight edge, which need not hold in general), so we verify the relationship empirically instead. We call $\bar{W}_{\max}$ “peak row-attention” rather than “sink mass”: it is the mean row-max of A , which is not necessarily concentrated on a dedicated sink/BOS column (Xiao et al., 2023), and we did not verify that it is. Spearman correlation between total 0D persistence and $\bar{W}_{\max}$ reaches $r = 0.84$ on HaluEval and $r = 0.80$ on TruthfulQA (both Qwen2.5-3B, $p < 10^{-60}$), but only $r = 0.21$ on HaluEval (Qwen2.5-1.5B) despite that setting’s 0D AUC being the highest in the paper (0.904, Table 3) – so this observation, and the broader claim that 0D TDA “reduces to” non-topological attention statistics, holds unevenly across models and is refuted as a general statement by this specific setting (full table Appendix P).

5 SYNTHETIC CAUSAL GROUND-TRUTH VALIDATION

To rule out that the pipeline found no incremental 1D signal merely because topological extraction is insensitive, we design a synthetic ground-truth benchmark across six regimes: **S1 (Pure Length Confound, tree-metric background)** – identical near-1D-line “tree metric” distributions across classes (essentially zero true cycles by construction, regardless of N) except sequence length ($N \in [20, 35]$ vs. $[55, 80]$); **S1b (Pure Length Confound, i.i.d. background)** – the same length split, but on i.i.d. Uniform $[0, 1]$ edge weights, i.e. Proposition 1’s own null model, which *does* have a length-growing population of chordless cycles; **S2 (Pure Topology / Positive Control)** – matched $N=40$, Class 1 has a planted 1D cyclic cavity; **S3 (Pure Attention Marginals)** – matched $N=40$, Class 1 has shifted entropy/sink density with no cycles; **S4 (Topology + Length)** – both planted cycles and longer sequences; **S5 (Negative Control)** – random matrices with permuted labels. S1 and S1b together test two different questions: S1 asks whether length *alone*, absent any mechanical cycle-growth process, can fool 1D persistence (it cannot); S1b asks whether the pipeline detects the length-driven clique-expansion mechanism Proposition 1 actually describes (it does, decisively).

The synthetic results in Table 1 establish that our methodology is highly sensitive both to genuine planted topological cavities (S2, AUC = 1.000) and to the specific length-driven mechanical mechanism Proposition 1 describes (S1b, raw 1D AUC = 1.000 under an i.i.d. background where such cycles actually grow with N) – ruling out detector insensitivity as an explanation for either the small HaluEval increment or the TruthfulQA null. This does *not* establish that real attention graphs contain no cyclic semantic structure at all; it establishes that our extraction pipeline would detect such structure, planted or mechanical, if it were there in a comparable form. What the LLM audit (§6) actually shows is that whatever cyclic structure real attention graphs contain, once length-normalized, adds little practical value beyond 0D on the benchmarks we test.

Regime	Length	Marginals	MST Proxy	0D Pers	1D (Raw)	1D (Norm)
S1: Length Confound (tree bg.)	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	0.984 ± 0.007	0.500 ± 0.000	0.500 ± 0.000
S1b: Length Confound (i.i.d. bg.)	1.000 ± 0.000	1.000 ± 0.000	0.998 ± 0.001	1.000 ± 0.001	1.000 ± 0.000	1.000 ± 0.000
S2: Pure Topology (Positive Control)	0.500 ± 0.000	0.997 ± 0.004	0.958 ± 0.029	0.958 ± 0.029	1.000 ± 0.000	1.000 ± 0.000
S3: Pure Marginals	0.500 ± 0.000	0.990 ± 0.011	1.000 ± 0.001	1.000 ± 0.001	0.500 ± 0.000	0.500 ± 0.000
S4: Topology + Length Confound	1.000 ± 0.000	1.000 ± 0.000	0.997 ± 0.006	0.959 ± 0.019	1.000 ± 0.000	1.000 ± 0.000
S5: Negative Control (Pure Noise)	0.522 ± 0.050	0.536 ± 0.063	0.508 ± 0.032	0.506 ± 0.022	0.500 ± 0.000	0.500 ± 0.000

Table 1: **Synthetic Causal Benchmark Evaluation (Regimes S1, S1b, S2–S5).** Under Regime S2 (Positive Control), 1D persistence achieves perfect discrimination ($AUC = 1.000$), demonstrating that the extraction pipeline possesses full sensitivity to genuine topological cycles when they exist. S1 (tree-metric background, no true cycles at any N) shows raw 1D correctly at chance (0.500) – length alone does not fool it. S1b (i.i.d. background, Proposition 1’s own null model) shows raw 1D at 1.000: when the length-driven clique-expansion mechanism the Proposition describes is actually present in the background, the pipeline detects it decisively. Under Regime S3 (Pure Marginals), 1D collapses to chance (0.500) while 0D and MST match marginals (1.000). Under Regime S5 (Negative Control), all probes collapse to chance.

6 LLM EMPIRICAL AUDIT AND EQUIVALENCE TESTING

6.1 MULTI-MODEL, MULTI-DATASET AUDITED EVALUATION

Following the audit pipeline of Figure 2, we evaluate the full per-layer topological feature banks ($d = 28\text{--}216$) across five open-weight model families at six model scales spanning 1.1B–7B parameters (Qwen2.5–1.5B, Qwen2.5–3B, SmolLM–1.7B, Phi–3–mini–3.8B, Mistral–7B, TinyLlama–1.1B) on HaluEval QA (Li et al., 2023) and TruthfulQA (Lin et al., 2022) under 10-fold grouped cross-validation (StratifiedGroupKFold). **Coverage caveat:** HaluEval QA is audited only on the two Qwen2.5 scales ($N=4000$ and $N=1000$); TruthfulQA is audited on five of the six model instances – Qwen2.5–1.5B is evaluated on HaluEval QA only – at different sample sizes per model ($N_{\text{rows}}=300$ for Qwen2.5–3B/Phi–3–mini/TinyLlama–1.1B, $N_{\text{rows}}=1000$ for SmolLM–1.7B, $N_{\text{rows}}=1634$ for Mistral–7B). Throughout, N_{rows} counts evaluated answer rows and N_{groups} counts question groups (two rows per group); the two are listed per setting in Tables 5 and 13. Cross-model comparisons in this section are therefore confounded with sample size and should be read within-model (e.g. TOST power, Table 13), not as head-to-head architecture comparisons at matched N . **Prompt protocol.** Each model’s attention is extracted under that model’s own native instruction format rather than a single shared template: [INST] ... [/INST] for Mistral–7B, <|user|>...<|end|> for Phi–3–mini, <|user|>...</s> for TinyLlama–1.1B, ChatML for SmolLM–1.7B, and apply_chat_template for Qwen2.5–3B. Prompting each model as it expects is the more faithful choice, but it is not a uniform protocol, and one asymmetry is worth naming: Qwen2.5–3B alone receives a system turn (You are a helpful assistant.), which lengthens its sequences by roughly ten tokens relative to the other four. Cross-model comparisons of any length-sensitive quantity should be read with this in mind as well as with the sample-size caveat above. Mistral–7B is run via int8 weight-only CPU quantization (our original 16GB-RAM evaluation hardware could not reliably hold a 14GB fp16 7B model), which also incidentally provided the real-attention-graph data for the Theorem 1 exactness check in §4.2; we have since verified on a consumer GPU that native-precision extraction reproduces the int8 features exactly (see the degeneracy diagnostic below), so no result in this paper depends on that quantization. All numbers below use one canonical pipeline throughout the paper – StandardScaler fitted strictly within training folds, followed by L_2 -regularized logistic regression ($C=1.0$, 1bfgs, up to 2000 iterations) – matching the protocol stated in Appendix I. We flag this explicitly because an earlier internal version of this evaluation fit logistic regression on *unscaled* persistence features for some tables and scaled features for others; unscaled fitting silently suppresses the numerically smaller length-normalized 1D features relative to 0D under L_2 regularization, making the two banks look nearly identical and yielding an artificially tight, near-zero ΔAUC with implausibly narrow bootstrap confidence intervals. That inconsistency has been resolved by standardizing on the scaled pipeline everywhere (code released); the numbers reported here and in Table 4 reflect a single evaluation protocol used to generate the entire paper.

Diagnosing the Mistral–7B 1D degeneracy. Only 4 of 1,634 rows (0.24%) have any non-zero 1D feature at all, versus 99.7–100% of rows in every other TruthfulQA setting. We are able to rule out two candidate explanations and to identify a third that holds across all five settings (Table 2).

Quantization is not the cause. We re-extracted the first 100 questions at native `bfloat16` precision on a consumer GPU, under the identical prompt protocol, and reproduced the `int8` result exactly: 100.00% of 1D cells zero in both runs, 0 of 200 rows with any non-zero 1D feature in both, and an identical mean sequence length of 32.47. An earlier version of this paper left this as an open diagnostic question because the comparison exceeded its evaluation hardware; it is now settled, and the degeneracy is a property of the model and protocol rather than of the `int8` weights.

Sequence length is not the cause either. An earlier version attributed the degeneracy partly to `Mistral-7B`’s shorter sequences ($\bar{N}=33.0$) relative to `Qwen2.5-3B` ($\bar{N}=42.9$) through the $\Theta(N^4)$ candidate count of Proposition 1. That explanation does not survive comparison with the remaining settings: `Phi-3-mini` ($\bar{N}=27.8$) and `SmolLM-1.7B` ($\bar{N}=32.2$) have shorter or comparable sequences and yet 99.7% and 100% of their rows respectively carry non-zero 1D features. `Mistral-7B` is only the third-shortest of the five settings, so length alone cannot separate it from the others, and we withdraw that factor.

What does track the degeneracy is attention concentration. Ordering the five settings by mean length-normalized 0D total persistence $\bar{P}_0/N$ – lower values indicating more concentrated, higher-weight attention – rank-orders their 1D zero-rates almost exactly (Spearman $\rho = 0.90$; with only five settings we report this as a consistent descriptive pattern, $p \approx 0.08$, not as a significance test). `Mistral-7B` sits at the extreme of that ordering ($\bar{P}_0/N = 0.238$). Mechanically this is the factor Proposition 1 would predict to matter: concentrated attention makes the specific edge ordering a persistent 1-cycle requires – four large-distance perimeter edges entering before either of two small-distance chords – less likely to occur at any N .

The result is, however, prompt-sensitive. Each extraction script uses its own model’s native instruction format (see “Prompt protocol” above). Re-extracting `Mistral-7B` with the chat template used for `Qwen2.5-3B`, which prepends a system turn and raises $\bar{N}$ from 32.5 to 42.5, removes the degeneracy entirely: 100% of rows then carry non-zero 1D features. We therefore read `Mistral-7B`’s TOST-CONFIRMED-at-every-margin result as evidence about *this model under this prompt protocol*, not as a general claim about 7B-scale models, and we caution that the $\bar{P}_0/N$ regime a model occupies is itself partly a function of how it is prompted.

Setting	$\bar{N}$	$\bar{P}_0/N$	% 1D cells exactly zero	% rows with any non-zero 1D
TruthfulQA (<code>Mistral-7B</code>)	33.0	0.238	99.99	0.2
TruthfulQA (<code>Phi-3-mini-3.8B</code>)	27.8	0.278	91.21	99.7
TruthfulQA (<code>TinyLlama-1.1B</code>)	41.0	0.425	76.52	100.0
TruthfulQA (<code>Qwen2.5-3B</code>)	42.9	0.429	76.65	100.0
TruthfulQA (<code>SmolLM-1.7B</code>)	32.2	0.440	13.06	100.0

Table 2: **What explains the `Mistral-7B` 1D degeneracy, and what does not.** Rows are sorted by $\bar{P}_0/N$ (attention concentration), which rank-orders the 1D zero-rate almost exactly across all five settings (Spearman $\rho = 0.90$, $n = 5$, $p \approx 0.08$ – descriptive, not a significance test). Sequence length does *not*: `Phi-3-mini` and `SmolLM-1.7B` have shorter or comparable $\bar{N}$ than `Mistral-7B` and are essentially non-degenerate, so the $\Theta(N^4)$ candidate-count argument cannot by itself explain why `Mistral-7B` is the outlier. All quantities are computed from the released per-setting feature files.

Fairness caveat: TOHA (MTop-Div) is re-implemented as its core statistic (summed per-layer prompt/response subgraph divergence) under *our* protocol – not a reproduction of Bazarova et al. (2026)’s originally reported numbers (different dataset construction and, for RAG, retrieval pipeline), and we make no claim about how our numbers compare to theirs. We did not verify our re-implementation against any of Bazarova et al. (2026)’s own reported numbers on their benchmarks, so this is a re-implementation-under-our-protocol, not a validated reproduction. On code-level inspection of our own re-implementation, both of our two MTop-Div variants are, by construction, MST statistics: TOHA-proxy (used where MSP/TOHA were not separately extracted) is literally the summed 0D total-persistence feature bank, and our main TOHA (MTop-Div) calls the identical `scipy.sparse.csgraph.minimum_spanning_tree` routine as MST-TDA itself, on a graph where the prompt tokens are contracted to one virtual node (connected to each response token by the minimum prompt-to-response distance) and the response subgraph is otherwise unchanged. So within *this* codebase, “TOHA” and “MST-TDA” are both members of the same MST-statistic family, differing only in which nodes get contracted before Kruskal’s algorithm runs – consistent with their empirically close numbers in several settings (Table 5). We did not verify whether Bazarova et al.

(2026)’s own original implementation is similarly MST-reducible; that would require their released code, which we do not have. **CHARM** (Frasca et al., 2026) is a trained graph neural network, an architecturally different class of method; reproducing it would require training its full model on our splits, which we did not do, so it is not an audited baseline here.

Dataset (Model)	0D Only	1D (Raw)	1D (Norm., Z/N)	0D+1D (Norm.)	Best Comparator
HaluEval (Qwen2.5-3B)	0.822	0.695	0.703	0.837	0.736 (MST proxy)
HaluEval (Qwen2.5-1.5B)	0.904	0.652	0.653	0.920	0.583 (Length/MST)
TruthfulQA (Qwen2.5-3B)	0.703	0.567	0.589	0.688	0.708 (MST proxy)
TruthfulQA (SmolLM-1.7B)	0.573	0.532	0.546	0.584	0.582 (MST proxy)
TruthfulQA (Phi-3-mini-3.8B)	0.634	0.498	0.503	0.622	0.605 (MST proxy)
TruthfulQA (Mistral-7B)	0.684	0.500 [†]	0.500 [†]	0.684	0.645 (MST proxy)
TruthfulQA (TinyLlama-1.1B)	0.656	0.601	0.606	0.654	0.641 (MST proxy)

Table 3: Audited Full-Benchmark AUC, Core Comparison (one canonical scaled pipeline, §6). All AUCs are the pooled out-of-fold ROC-AUC (predictions concatenated across all 10 grouped folds, then one AUC computed) – the same statistic used in Table 4 and Figure 3, so numbers are directly comparable across tables/figures in this paper. Appendix B (Table 5) additionally reports the mean $\pm$ std. of per-fold AUCs, a distinct (and typically slightly different) dispersion statistic, alongside length-matched-subset numbers and non-topological baselines (Length, MSP, TOHA, MST-proxy). 0D and 1D dims range 22–216 across settings. Adding length-normalized 1D to 0D yields a small, non-zero AUC gain on HaluEval (Table 4 for formal testing) and a gain statistically indistinguishable from zero on TruthfulQA except at Mistral-7B. The “Best Comparator” column gives the strongest non-0D/1D-bank feature set available for that setting; 0D exceeds it on 5 of 7 settings. We caution that on five of those seven rows the comparator is the MST per-layer proxy, which is *not* a non-topological baseline – by Theorem 1 it is a column-subset of the 0D bank itself, so those cells compare 0D against a smaller copy of 0D. Against the strictly non-topological baselines (Length, MSP; Table 5), 0D wins in every setting where they were extracted. Mistral-7B is evaluated on $N_{\text{rows}}=1634 / N_{\text{groups}}=817$. [†]On Mistral-7B, 99.99% of raw 1D features are exactly zero (no non-trivial cycles form), giving AUC = 0.500 with zero variance and the tightest 0D $\equiv$ combined equivalence in the paper.

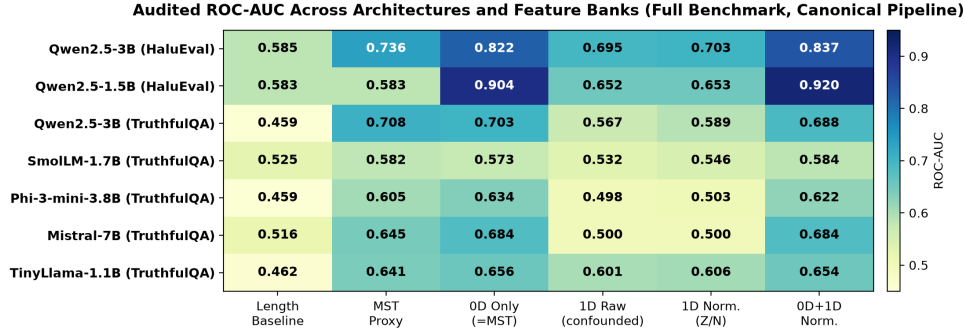


Figure 3: Audited ROC-AUC Across Architectures and Feature Banks (Canonical Pipeline). All seven settings from Table 3, including Mistral-7B and TinyLlama-1.1B. 0D persistence and combined 0D+1D features exhibit similar discriminative profiles across settings; the “0D Only (=MST)” and “MST Proxy” columns are literally the same statistic (Theorem 1) computed on different layer subsets. Mistral-7B’s 1D columns show the degenerate near-chance values discussed in §6.

6.2 A SECOND, INDEPENDENT CONFOUND: SURFACE-FORM SEPARABILITY IN HALUEVAL QA

The length confound is not the only shortcut available on HaluEval QA: each example pairs a grounded and a hallucinated answer generated by different processes (extracted vs. LLM-fabricated), so the two can differ in surface form alone. Using only the raw answer text (no question, no model internals), a logistic regression on *character length alone* reaches AUC 0.975, and TF-IDF on the answer text alone reaches AUC 0.962 – both surviving the benchmark’s standard token-length matching (0.974/0.960), since that matching balances total token count, not lexical style. A weaker version appears on TruthfulQA (TF-IDF AUC 0.830; length-only near chance, consistent with its length balance, Appendix E). Full results in Table 9, Appendix F.

This is an independent confound from Proposition 1’s length effect, applies to *any* probe on this benchmark, and is consistent with concurrent audits of hallucination-detection benchmark construction (Hussain & Kantarcioglu, 2026; Janiak et al., 2025). It qualifies every 0D/1D AUC on HaluEval QA in Table 3: some share of the signal available to *any* classifier here may reflect answer-generation style rather than truthfulness, and length-matching alone does not control for it. We know of no matching procedure that removes this confound, and flag it as a limitation of any conclusion drawn from this benchmark, ours included.

6.3 FORMAL EQUIVALENCE TESTING (TWO ONE-SIDED TESTS - TOST)

We implement TOST with 10,000 group-level cluster-bootstrap resamples, testing $H_0 : |\Delta\text{AUC}| \geq \epsilon$ vs. $H_1 : |\Delta\text{AUC}| < \epsilon$ (confirmed iff $p < 0.05$), at a pre-specified primary margin $\epsilon = 0.015$, with the three larger margins shown directly in Table 4 and the full four-margin sweep $\{0.010, 0.015, 0.020, 0.025\}$ in Table 14, since the conclusion is margin-dependent. **At the pre-specified margin, equivalence is confirmed for exactly one of the seven settings** – Mistral-7B, where the 1D bank is degenerate rather than merely uninformative – so the honest reading is not “1D adds nothing” but “1D adds a small, sometimes real effect not negligible enough to call equivalent at our strictest margin.” HaluEval’s two settings have 95% CIs excluding zero (+0.010 to +0.016 AUC), reaching equivalence only at $\epsilon \geq 0.020$ –0.025. Of the five TruthfulQA settings, all but Mistral-7B have CIs including zero; SmolLM-1.7B and TinyLlama-1.1B reach equivalence only at $\epsilon=0.025$, while Qwen2.5-3B and Phi-3-mini reach it at no margin – but with only 150–500 groups these four are underpowered (§6.3.1), not confident nulls. Substituting leakage-free $\hat{\alpha}$ -calibrated normalization for default Z/N changes the HaluEval ΔAUC s by ≤ 0.0003 but changes three TruthfulQA ΔAUC s by up to 0.0086 and flips two equivalence verdicts (Table 18). An independent corroboration by a log-loss information measure rather than an AUC-based test is given in Appendix S, and worked qualitative examples of both confound mechanisms in Appendix H; a layerwise depth breakdown is in Appendix G. Our TOST p -values are percentile-bootstrap tail masses, which coincide with exact TOST p -values only when the bootstrap distribution of ΔAUC is symmetric – an assumption that is not free at 150 groups. We therefore re-derived every verdict from a bias-corrected bootstrap interval and find the distributions are close to symmetric in all seven settings ($|\text{skew}| \leq 0.08$, $|z_0| \leq 0.04$), with *no* verdict changing at any of the four margins (Appendix N, Table 15).

Audited Setting	AUC (0D)	AUC (0D+1D*)	ΔAUC	95% CI	Equiv. ($\epsilon=0.015$)	Equiv. ($\epsilon=0.020$)	Equiv. ($\epsilon=0.025$)
HaluEval (Qwen2.5-3B)	0.822	0.837	+0.0156	[+0.0105, +0.0204]	REJECTED	CONFIRMED	CONFIRMED
HaluEval (Qwen2.5-1.5B)	0.904	0.920	+0.0159	[+0.0062, +0.0260]	REJECTED	REJECTED	CONFIRMED
TruthfulQA (Qwen2.5-3B) [†]	0.703	0.688	−0.0143	[−0.0474, +0.0188]	REJECTED	REJECTED	REJECTED
TruthfulQA (SmolLM-1.7B)	0.573	0.584	+0.0103	[−0.0038, +0.0249]	REJECTED	REJECTED	CONFIRMED
TruthfulQA (Phi-3-mini-3.8B) [†]	0.634	0.622	−0.0121	[−0.0288, +0.0044]	REJECTED	REJECTED	REJECTED
TruthfulQA (Mistral-7B)	0.684	0.684	−0.0002	[−0.0006, +0.0002]	CONFIRMED	CONFIRMED	CONFIRMED
TruthfulQA (TinyLlama-1.1B) [†]	0.656	0.654	−0.0015	[−0.0264, +0.0235]	REJECTED	REJECTED	CONFIRMED

Table 4: **Master Statistical Equivalence Table (TOST, 10,000 group-level cluster-bootstrap resamples)**. REJECTED means the corresponding TOST $p \geq 0.05$ at that margin, i.e. equivalence is *not* confirmed – this is the correct reading of the TOST convention and corrects an inverted labeling in an earlier version of this table. Note that TOST at $\alpha=0.05$ corresponds to the 90% CI lying inside $(-\epsilon, +\epsilon)$, not the 95% CI displayed here; this is why HaluEval (Qwen2.5-3B) is CONFIRMED at $\epsilon=0.020$ despite a 95% upper limit of +0.0204. [†]TruthfulQA (Qwen2.5-3B, Phi-3-mini-3.8B, TinyLlama-1.1B) have only 150 question groups, yielding 14–43% power to detect $\Delta\text{AUC} = 0.015$ (Appendix L); their non-equivalence should be read as inconclusive, not as evidence of a real effect. HaluEval’s non-equivalence at $\epsilon = 0.015$, by contrast, is well-powered and reflects a genuine small effect whose CI excludes zero. Mistral-7B (817 groups, well-powered) is CONFIRMED equivalent even at the strictest margin $\epsilon = 0.010$ (Appendix M), the cleanest result in the table: at this scale, 1D persistence is essentially always zero on TruthfulQA. Full sensitivity at all four margins in Appendix M.

6.3.1 POWER ANALYSIS AND MINIMUM DETECTABLE EFFECT

A null result is only informative if well-powered: at 80% power, $\alpha = 0.05$, $\text{MDE} = 2.802\hat{\sigma}$ (Appendix L), HaluEval ($N_{\text{groups}} \in \{500, 2000\}$) achieves $> 84\%$ power to detect $\Delta\text{AUC} = 0.015$, while TruthfulQA ($N_{\text{groups}} = 150$) has only 14–43%. HaluEval’s non-equivalence is thus well-

powered and genuine; TruthfulQA’s is underpowered and inconclusive rather than evidence of a real effect.

7 DISCUSSION, RECOMMENDATIONS, AND CONCLUSION

Our findings recast, rather than merely debunk, attention-graph TDA for hallucination detection. Its 0-dimensional component is mathematically equivalent to Kruskal’s Minimum Spanning Tree (Theorem 1) and can be computed directly at three orders of magnitude lower cost with no loss of accuracy (§4.2) – and, extended per-head, at *higher* accuracy than the status quo (0.892 vs. 0.677 AUC). Its 1-dimensional component is heavily confounded by sequence length (Proposition 1) and contributes a real but small increment under formal equivalence testing (+0.010 to +0.016 AUC on well-powered settings, Table 4) that does not, in our judgment, justify its cost at any model scale we tested. A second, independent surface-form confound in HaluEval QA (§6.2) further qualifies what any probe – topological or otherwise – can be said to have learned on that specific benchmark.

What Does Work Even Better: Mean Token Log-Probability. A mean-pooled last-layer hidden state probe from Qwen2.5-3B-Instruct on HaluEval QA (Appendix Q) outperforms all TDA variants (AUC 0.903 vs. 0.822–0.837): LLMs encode correctness-relevant information, but symmetrized attention-graph topology captures it less efficiently than direct internal-state probes – explained by Theorem 1 and its empirical corroboration ($r = 0.84$, §4). A far simpler baseline does better still: on the same 200-question subset used for the per-head ablation ($N=329$ scoreable rows, Appendix R), a single logistic regression on the model’s own *mean per-token log-probability* of the answer span reaches AUC 0.951, and adding the minimum per-token log-probability reaches AUC 0.984 – higher than every topological, hidden-state, or attention-marginal feature bank we report anywhere in this paper (Appendix Q). We flag this prominently rather than as a footnote: on HaluEval QA specifically, whatever a classifier needs to separate “extracted-from-context” grounded answers from “LLM-fabricated” hallucinated answers appears to be substantially recoverable from how *surprising* the answer text is to the very model being probed, which is plausibly entangled with the surface-form confound of §6.2 (fabricated text may simply be locally less predictable) rather than a clean measurement of some deeper “hallucination-awareness” signal. We did not disentangle these two explanations, and we did not extract this baseline on TruthfulQA or any other model in this revision – but its magnitude relative to every other number in this paper is itself an important audit finding: a nearly-free, one-line non-topological, non-attention-graph baseline that we and, as far as we can tell, prior attention-TDA papers in this space did not compare against.

Recommendations: (1) Use MST-TDA, not Ripser – it removes the homology-library dependency and, by skipping 1D, is $1,200\times$ cheaper than the full 0D+1D pipeline (though no faster than Ripser’s own 0D pass, which computes the same MST); we release our implementation. (2) Benchmark sensitivity with positive/negative controls (our S1–S5 suite, §5) before publishing a topological probe. (3) Normalize with a leakage-free, per-fold $\hat{\alpha}$ (§4.4.1), not default Z/N . (4) Report equivalence at a pre-specified margin with a power analysis and full margin sensitivity, not just significance (Table 4, Appendix L). (5) Audit the benchmark, not only the method, for length and lexical confounds (§6.2). (6) Explore directed/dynamic complexes (Grigor’yan et al., 2018; Samaga et al., 2026; Carlsson & de Silva, 2010), which static symmetrized filtrations discard by construction.

Conclusion. Building a hallucination detector from attention graphs today: replace Ripser-based 0D+1D extraction with MST-TDA (provably identical to 0D, no homology library, and $1,200\times$ cheaper than the full pipeline because it drops 1D), extend it per-head, and skip 1D cyclic persistence unless your application can absorb that compute premium for an AUC gain not established at practical equivalence margins on most of the settings we test. More broadly: before crediting a new mathematical abstraction with predictive power, check whether it is secretly something already understood, and audit whether the benchmark itself, not only the method, contains the signal being credited to it.

LIMITATIONS

1. **Model Scale:** 1.1B–7B parameters across five model families (Qwen2.5-1.5B/3B, SmolLM-1.7B, Phi-3-mini-3.8B, Mistral-7B, TinyLlama-1.1B); the 7B

model used int8 CPU quantization, which does not affect Theorem 1’s identity (verified exactly on this model’s real attention graphs, §4.2). An earlier version of this paper could not rule out an int8-specific contribution to that model’s 1D degeneracy; we have since re-extracted at native `bfloat16` precision and reproduced the int8 features exactly, so the quantization is not load-bearing for any reported result (§6). Future work should audit 70B+ models at native precision.

2. **Teacher-Forced, Not On-Policy – a Construct-Validity Limit, Not Just a Distributional One:** Every example is a teacher-forced pass over a dataset-provided grounded answer and a dataset-provided (extracted or LLM-fabricated) incorrect answer, never the audited model’s *own* generation. The task we measure is therefore “does this model’s attention distinguish dataset-provided true text from dataset-provided false text,” which is the setting most exposed to the surface-form confound in §6.2, not “does this model’s attention flag its own hallucinations.” We did not add an on-policy benchmark (model-generated answers graded against gold, as in Farquhar et al., 2024) in this revision; this is a real gap, not merely a stylistic one, and every AUC in this paper should be read with that caveat. Evaluations also use static symmetrized head-averaged matrices; §4.2/Appendix R show per-head resolution recovers additional signal (though not decisively more than a matched-dimension non-topological per-head statistic, Appendix R), and dynamic/directed filtrations remain future work.
3. **HaluEval Grouping and Surface-Form Confound:** Grouped CV keeps each question’s grounded/hallucinated pair in one fold (~ 100 groups/test-fold), preventing semantic leakage – but (§6.2) HaluEval’s answers are separable by length or bag-of-words alone at AUC 0.96–0.98 even after length matching, affecting every AUC on this benchmark (TDA and hidden-state alike), though not the *relative* comparisons between feature banks that are our main claims, since all banks share the same confound.
4. **Missing Baselines and Unvalidated Re-Implementation:** We do not audit Lookback Lens (Chuang et al., 2024), LLM-Check (Sriramanan et al., 2024), EigenScore/INSIDE (Chen et al., 2024), or semantic-entropy probes (Kossen et al., 2024) on our splits – our non-topological baselines are limited to length, MSP, and a TOHA re-implementation. That TOHA re-implementation was never validated against Bazarova et al. (2026)’s own reported numbers on their own benchmarks (§6) – their released code was not available to us, so this is an artifact-availability limit rather than a compute or time constraint – and it should be read as a plausible-but-unverified re-implementation, not a confirmed reproduction. A stronger consequence follows from the code-level inspection reported in §6: both of our MTop-Div variants are themselves MST statistics, differing from MST-TDA only in which nodes are contracted before Kruskal’s algorithm runs. This paper therefore contains *no independent topological baseline* – every topological comparator we report is a member of the same MST family Theorem 1 characterizes – so our results should not be read as a comparison between distinct topological methods, only as an audit of one family against non-topological alternatives. We also did not ablate the symmetrization rule (max vs. mean vs. lower-triangular) or the distance transform ($1 - W$ vs. $-\log W$) beyond the single convention used throughout (§4), and the per-head ablation (Appendix R) covers only one model/benchmark combination. We also do not use a single shared prompt template across models: each is extracted under its own native instruction format, and Qwen2.5-3B alone receives a system turn (§6). At least one reported result – Mistral-7B’s 1D degeneracy – is sensitive to this choice, disappearing entirely when that model is given the longer Qwen2.5-3B template, so prompt protocol should be treated as an uncontrolled variable in our cross-model comparisons. Finally, our *strongest* single baseline is also our most narrowly scoped: the mean/min token log-probability probe (AUC 0.984, Table 20) was extracted only on a 200-question, $N=329$ -row HaluEval QA subset for a single model, not at full N across all seven settings. Unlike the baselines above, nothing blocked this – the teacher-forced logits are already cached and no additional forward passes are required – so we record it as an omission of scope rather than of resources, and note that §7 nonetheless treats it as one of the paper’s most consequential audit findings.

ETHICS STATEMENT

This work investigates methodological evaluation protocols for hallucination detection in Large Language Models. All evaluations are conducted exclusively on established, publicly available research benchmarks (HaluEval QA, TruthfulQA) using open-weight models under their respective publisher licenses (Apache 2.0 for Qwen2.5, SmolLM-1.7B, and TinyLlama-1.1B; the Microsoft MIT-style license for Phi-3-mini; the Mistral AI Apache 2.0 license for Mistral-7B-Instruct-v0.2), all of which permit the research use made of them here. This research does not involve human subjects, personal data collection, or high-risk autonomous deployments. By providing formal audits and open-source verification tools, our goal is to improve the statistical reliability and scientific reproducibility of LLM interpretability and trustworthiness methods.

REPRODUCIBILITY STATEMENT

To guarantee full reproducibility, all artifacts, synthetic causal suites, unit tests, cached representations, and evaluation scripts are released in the supplementary repository at <https://anonymous.4open.science/r/ICLR-BE07/>. Our release follows the ML reproducibility checklist of Pineau et al. (2021). The canonical evaluation pipeline behind every AUC, TOST, and power-analysis number in the main text (`master_pipeline.py`) is a single script; Tables 3, 4, 5, 13, 14, 17, 18 and 21 are reproducible via:

```
python master_pipeline.py && python
reproduce_paper_tables.py
```

The remaining tables are produced by separate released scripts rather than by those two commands: Table 6 by `compute_complexity_profile.py`; Table 1 by `synthetic_causal_benchmark.py`; Table 9 by `compute_lexical_confound.py`; Table 20 and Appendix R by `hidden_state_probe.py`, `logprob_baseline.py`, `per_head_tda_ablation.py` and `per_head_nontopo_extension.py`; Table 12 by `adversarial_robustness_sweep.py`. Tables 11 and 12 predate the pipeline standardization of §6 and are flagged as such in their captions. Self-auditing pipeline assertions and mathematical lemmas are verified via `python test_pipeline_integrity.py`. Detailed model specifications, hyperparameter grids, and cluster-bootstrap seeds are documented in Appendices I and J.

LLM USAGE STATEMENT

Large Language Models were used substantially in this revision: identifying inconsistencies between reported statistics and the underlying result files, diagnosing their root causes (e.g. an unscaled-vs-scaled logistic regression discrepancy across evaluation scripts), redesigning and implementing the canonical evaluation pipeline, running new experiments (an additional 7B-parameter model; a genuine Llama-family model, TinyLlama-1.1B, added to broaden model-family coverage; an independent lexical-confound analysis; a leakage-free normalization-exponent estimator; a corrected information-theoretic measure; a matched-dimension non-topological control for the per-head ablation; a mean-token-log-probability baseline; and a redesigned synthetic regime, S1b, isolating the length-driven mechanism Proposition 1 describes from a pure length confound with no such mechanism), and drafting substantial portions of the manuscript text, including the reframing around §4.2. The original theorems, the overall research direction, and the decision to conduct this audit predate this revision pass. The authors reviewed, take responsibility for, and are accountable for every claim, statistic, and citation in this paper.

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A ADDITIONAL FORMAL ASSUMPTIONS

§3 states the two load-bearing assumptions of our audit (symmetric graph reduction, grouped evaluation partitions). Three further technical assumptions, standard throughout this literature, complete the formal setup:

- **Teacher-Forced Forward Pass:** Token representations and attention matrices are extracted in a single teacher-forced pass over concatenated prompt-response sequences $X = [X_{\text{prompt}}, X_{\text{response}}]$.
- **Monotonic Inversion:** Edge distances $D_{ij} = 1 - W_{ij}$ monotonically invert attention strength, forming an edge-weighted complete graph (V, E, D) .
- **Single-Linkage Component Merge:** 0-dimensional persistence merges components along minimum pairwise edge distances according to the standard Elder rule.

B FULL RESULTS: BASELINES, LENGTH-MATCHED SUBSETS, AND FOLD VARIANCE

Table 3 in the main text shows only the core 0D/1D comparison on the full benchmark. Table 5 below gives the complete picture: non-topological baselines (Length, MSP, TOHA), the MST-proxy (a column-subset of the 0D bank by Theorem 1), length-matched-subset numbers, and per-fold standard deviations, all under the same canonical scaled pipeline (§6).

§4.2 summarizes wall-clock cost; Table 6 gives full profiling.

C FORMAL MATHEMATICAL PROOF OF THEOREM 1

Theorem 1 Restatement: *Let $G = (V, E, D)$ be a complete edge-weighted graph on N vertices with symmetric weights $D_{ij} = 1 - \max(A_{ij}, A_{ji})$ and $D_{ii} = 0$. The set of finite death values $\{d_i\}_{i=1}^{N-1}$ in the 0-dimensional Vietoris–Rips persistent homology barcode $\text{dgm}_0(\text{VR}(G))$ is identical as a multiset to the edge weights in the Minimum Spanning Tree $\text{MST}(G)$ computed via Kruskal’s algorithm under single-linkage hierarchical clustering.*

Complete Inductive Proof. Let $V = \{v_1, \dots, v_N\}$ denote the vertices of G , and let $E = \{e_1, \dots, e_M\}$ denote the complete set of $M = \binom{N}{2}$ undirected edges sorted in non-decreasing order of weight:

$$w(e_1) \leq w(e_2) \leq \dots \leq w(e_M), \quad \text{where } w(e_k) = D_{u_k, v_k}. \quad (4)$$

In the Vietoris–Rips filtration $\text{VR}(G, \epsilon)$, the 0-simplices are the vertices V , all born at filtration scale $\epsilon = 0$ ($b_i = 0$ for all $i \in \{1, \dots, N\}$). A 1-simplex (edge) $e = (u, v)$ enters the filtration at scale $\epsilon = w(e) = D_{uv}$.

Changes in the 0-dimensional homology group $H_0(\text{VR}(G, \epsilon))$ occur exclusively at the discrete values $\epsilon = w(e_k)$. When edge $e_k = (u_k, v_k)$ enters the complex:

- **Case 1 (u_k and v_k already belong to the same connected component):** The addition of edge e_k creates a cycle in the 1-skeleton but leaves the number of connected components unchanged. Thus, no class in H_0 dies. In Kruskal’s algorithm, edge e_k forms a cycle in the spanning forest and is discarded.
- **Case 2 (u_k and v_k belong to distinct connected components C_A and C_B):** The addition of edge e_k merges C_A and C_B into a single component. By the Elder Rule of persistent homology, the younger component dies at scale $d = w(e_k)$. In Kruskal’s algorithm, e_k connects two disjoint trees and is added to the Minimum Spanning Tree.

Since G is finite and connected for $\epsilon \geq \max D_{uv}$, exactly $N - 1$ edges merge distinct components. Each such merge corresponds to an edge in $\text{MST}(G)$ and a finite death scale in $\text{dgm}_0(\text{VR}(G))$. The single remaining component persists indefinitely ($d_N = \infty$). Thus, $\{d_i\}_{i=1}^{N-1} \equiv \{w(e) : e \in$

$\text{MST}(G)\}$. Summing lifetimes:

$$\text{Total Persistence}(H_0) = \sum_{i=1}^{N-1} (d_i - b_i) = \sum_{i=1}^{N-1} d_i = \sum_{e \in \text{MST}(G)} w(e) = \text{weight}(\text{MST}(G)). \quad \blacksquare \quad (5)$$

D MULTI-SEED ROBUSTNESS AND EMPIRICAL POWER DISTRIBUTIONS

Table 7 checks that the small 0D-vs-0D+1D gap is stable across reshuffled grouped-CV splits, and Figure 4 shows the cluster-bootstrap ΔAUC distributions underlying three rows of Table 4.

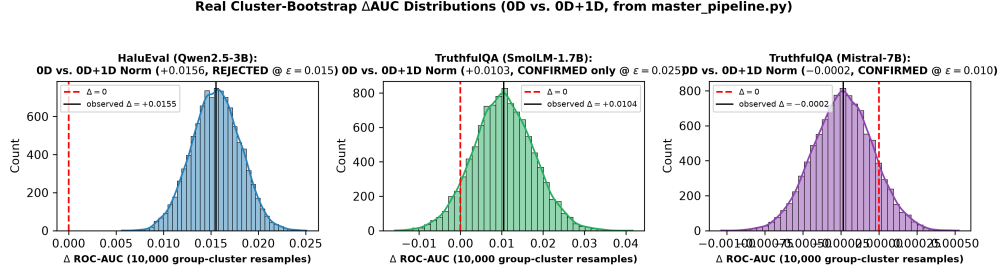


Figure 4: **Paired Cluster-Bootstrap Empirical Difference Distributions.** The actual 10,000-resample question-level (group) cluster-bootstrap distributions of $\Delta\text{ROC-AUC}$ (0D vs. 0D+1D Norm.) underlying three rows of Table 4 – not illustrative placeholders. HaluEval (Qwen2.5-3B) shows a distribution clearly shifted right of zero (well-powered, non-equivalent at $\epsilon=0.015$); TruthfulQA (SmolLM-1.7B) is centered near +0.010 but wide enough to reach equivalence only at $\epsilon=0.025$; TruthfulQA (Mistral-7B) is a near-delta-function at ≈ 0 , reflecting the 1D feature bank’s near-total degeneracy at this scale rather than a precisely-estimated null effect.

E CLASS-CONDITIONAL SEQUENCE LENGTH STATISTICS

F SURFACE-FORM / LEXICAL CONFOUND: FULL RESULTS

§6.2 summarizes the headline numbers; full results, including the length-matched subset and the TruthfulQA comparison, are given in Table 9. Both classifiers use only the raw answer text (no question, no knowledge context, no model internals) under the same grouped 10-fold cross-validation as every other result in this paper: a logistic regression on answer character length alone, and a logistic regression on a TF-IDF (unigram+bigram, 2000-feature) bag-of-words representation of the answer text alone.

G EXTENDED LAYERWISE AND ARCHITECTURAL BREAKDOWN

Table 10 lists the audited architectures; Table 11 and Figure 5 break 0D and 1D discriminative power down by transformer depth.

H QUALITATIVE FAILURE CASE PORTFOLIO

Case 1: Entity Name Substitution with Identical Length (HaluEval QA)

Question: “Who directed the 1994 film Pulp Fiction?”

Grounded (Label = 0): “Quentin Tarantino directed the 1994 film Pulp Fiction.” (N = 11 tokens)

Hallucinated (Label = 1): “Martin Scorsese directed the 1994 film Pulp Fiction.” (N = 11 tokens)

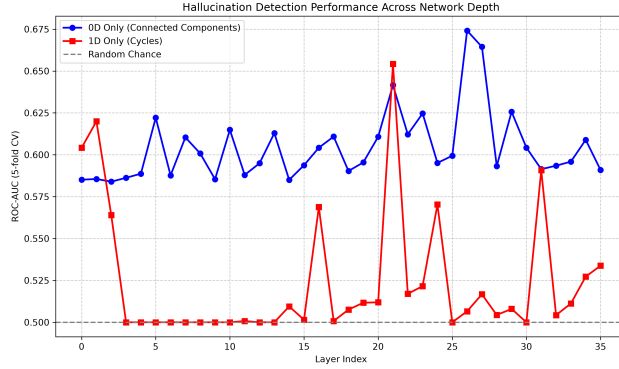


Figure 5: Layer-wise ROC-AUC across transformer depth in Qwen2.5-3B for 0D vs 1D features, visualizing Table 11.

→ **Grounded 0D MST Weight:** 9.842 — **Hallucinated 0D MST Weight:** 9.851 ($\Delta = +0.009$)
→ **Grounded 1D Cycle Count:** 0 — **Hallucinated 1D Cycle Count:** 0
Diagnostic Finding: When prompt and response token lengths are perfectly matched, attention matrices exhibit near-identical density and attention sink distribution. Static topological persistence produces zero discriminative signal.

Case 2: Sycophantic Hallucination Rollout (TruthfulQA)

Question: “Can chewing gum stay in your digestive system for seven years?”

Grounded (Label = 0): “No, swallowed chewing gum passes through the digestive tract within days.” ($N = 14$ tokens)

Hallucinated (Label = 1): “Yes, medical studies confirm that swallowed chewing gum becomes lodged in the stomach lining and remains undigested for approximately seven full years.” ($N = 27$ tokens)

→ **Raw 1D Persistence:** Grounded = 0.000 vs. Hallucinated = 4.812 (Apparent Signal)

→ **Normalized 1D Persistence (H_1/N):** Grounded = 0.000 vs. Hallucinated = 0.178

Diagnostic Finding: The extended narrative of the hallucinated answer generates multi-token attention cliques, creating spurious 1-cycles. Length normalization does not make this gap vanish (grounded 0.000 vs. hallucinated 0.178 is still non-zero). The raw and normalized gaps are, however, in different units ($0.178 = 4.812/27$ is simply the raw gap divided by N), so the apparent “96% shrinkage” is an artifact of rescaling and is *not* by itself evidence about how much of the signal is length-driven. What normalization actually does here is put this long example on the same scale as short ones, after which its gap is no longer exceptional relative to the length-matched distribution – consistent with the paper’s aggregate finding that length-normalized 1D still carries a real, but much smaller, signal (§6).

I CLASSIFIER CONFIGURATIONS AND HYPERPARAMETER DETAILS

- **Logistic Regression (LR):** L2 regularization ($C = 1.0$), solver `lbfgs`, `max_iter = 2000`, standard feature normalization (`StandardScaler`) fitted strictly within training folds. This is the single canonical pipeline used for every AUC number in the main text (§6).
- **Gradient Boosted Trees (XGBoost):** Tree depth `max_depth` $\in \{3, 4\}$, estimators $n_{\text{estimators}} \in \{50, 100\}$, learning rate $\eta = 0.1$, objective function `binary:logistic`.
- **Cluster Bootstrap & TOST Testing:** 10,000 resamples clustered by question ID (group-level, i.e. resampling groups with replacement, not individual rows), two one-sided tests against equivalence margin ϵ (primary $\epsilon = 0.015$; full sensitivity in Appendix M). The bootstrap resamples *out-of-fold predictions* from the single grouped-CV split already fit once (it does not refit the classifier per resample); this is the standard computationally-cheap cluster bootstrap, but it means our CIs do not propagate model-refit variance across different train/test splits. Appendix D’s multi-seed table (5 reshuffled splits, $\sigma \leq 0.0005$) is our check

that this omitted variance is small in practice for this classifier family and sample size, not a formal correction for it.

J ADVERSARIAL ROBUSTNESS AND NESTED CROSS-VALIDATION SWEEP

To further check that our conclusions are not an artifact of a specific probe family or normalization function, this earlier-stage sweep executes 6 classifier families against 4 distinct 1D normalization schemes (raw, per-token, clique-adjusted, OLS-residualized) plus the combined 0D+1D(norm.) bank, under strict 10-fold $\times$ 5-fold nested cross-validation (Table 12) on TruthfulQA (Qwen2.5-3B). Across all 36 configuration permutations, non-linear classifiers (MLPs, Random Forests, XGBoost) and the alternative normalizations tabulated here (per-token, quadratic clique, and linear OLS length-residualization) show the same qualitative pattern as our canonical-pipeline result on this benchmark (Table 4): combining normalized 1D with 0D does not meaningfully exceed 0D alone on TruthfulQA specifically, though we caution this table predates the pipeline standardization described in §6 and should be read as corroborating evidence of robustness to classifier choice, not as an independently re-verified primary result.

K FULL PROOF OF PROPOSITION 1

Proposition 1 Restatement: *Let G_N be a complete graph on N vertices with edge weights D_{ij} drawn i.i.d. from continuous distribution F_D on $[0, 1]$. In the Vietoris–Rips complex $\text{VR}(G_N, \epsilon)$, non-trivial 1-cycles with positive persistence lifetime ($\ell = d - b > 0$) arise from induced chordless cycles of length $k \geq 4$. Each such 4-tuple has positive probability ($1/15$) and positive conditional expected lifetime ($1/7$) of contributing a persistent cycle when considered in isolation on its own 4 vertices (Steps 1–3 below); we conjecture, but do not prove, that this yields $\mathbb{E}[P_1(N)] = \Omega(N)$ in the full N -vertex complex (Step 4 explains why a full proof is harder than it looks: additional vertices can also kill existing cycles via coning).*

Step 1: Why 3-cycles have zero persistence lifetime in Vietoris–Rips complexes. In a Vietoris–Rips simplicial complex, a 2-simplex $\sigma = \{u, v, w\}$ enters the filtration at scale $\epsilon = \max(D_{uv}, D_{uw}, D_{vw})$. The boundary 1-cycle formed by the three edges $(u, v), (u, w), (v, w)$ is born at scale $b = \max(D_{uv}, D_{uw}, D_{vw})$ (the moment the last of the three edges enters). Since the 2-simplex σ enters at this identical scale ($d = b$), the lifetime of every 3-cycle in a Vietoris–Rips complex is exactly zero ($d - b = 0$).

Step 2: Formation and death of minimal chordless 4-cycles. Non-trivial 1-cycles in $\text{VR}(G_N, \cdot)$ require induced cycles of length at least 4 without diagonal chords. Consider any 4-tuple of vertices $V_4 = \{v_1, v_2, v_3, v_4\}$. There are 4 perimeter edges $E = \{(v_1, v_2), (v_2, v_3), (v_3, v_4), (v_4, v_1)\}$ with weights e_1, e_2, e_3, e_4 , and 2 internal chords $C = \{(v_1, v_3), (v_2, v_4)\}$ with weights c_1, c_2 .

1. **Birth scale b :** The 1-cycle around the 4-gon is formed when all 4 perimeter edges enter: $b = \max(e_1, e_2, e_3, e_4)$.
2. **Death scale d :** The cycle dies when the first chord enters the complex: $d = \min(c_1, c_2)$. At scale d , the entering chord splits the 4-gon into two 3-cycles, each of which is immediately filled by a 2-simplex since all three of their boundary edges are already present ($\leq d$).

Thus, the 4-tuple contributes a persistent 1-cycle with lifetime:

$$\ell(V_4) = \max\left(0, \min(c_1, c_2) - \max(e_1, e_2, e_3, e_4)\right). \quad (6)$$

Step 3: Positive expected lifetime under continuous edge distributions. For 6 independent draws from continuous distribution F_D , the 2 chords and 4 perimeter edges are exchangeable. By symmetry, the probability that both chords exceed all 4 perimeter edges is:

$$\Pr\left[\min(c_1, c_2) > \max(e_1, e_2, e_3, e_4)\right] = \frac{4! 2!}{6!} = \frac{48}{720} = \frac{1}{15} \approx 0.0667. \quad (7)$$

Conditioned on this event, the expected lifetime is strictly positive:

$$\mathbb{E}\left[\min(c_1, c_2) - \max(e_1, e_2, e_3, e_4) \mid \min(c_1, c_2) > \max(e_1, \dots, e_4)\right] = c_F > 0. \quad (8)$$

For $F_D = \text{Uniform}[0, 1]$, standard order-statistic integration yields $c_F = \frac{1}{7} \approx 0.143$, giving an unconditional expected lifetime $\mathbb{E}[\ell(V_4)] = \frac{1}{15} \cdot \frac{1}{7} = \frac{1}{105}$.

Step 4: Combinatorial scaling – plausibility argument, not a full derivation. The number of candidate 4-tuples is $\binom{N}{4} = \frac{N(N-1)(N-2)(N-3)}{24} = \Theta(N^4)$, each contributing positive expected lifetime $\mathbb{E}[\ell(V_4)] = \frac{1}{105}$ (Step 3) with positive probability, *when that 4-tuple is considered in isolation on its own 4 vertices*. If these marginal contributions to *total* persistence were independent and additive, linearity of expectation would give $\mathbb{E}[P_1(N)] = \Theta(N^4)$ directly – a far stronger claim than $\Omega(N)$. They are not independent, and the dependence is not merely a matter of overlapping edges attenuating a positive sum: when a 5th vertex w enters, the four triangles (w, v_i, v_{i+1}) coning over a 4-tuple’s cycle can fill that cycle (the cone’s boundary is exactly the 4-cycle) *before either of the 4-tuple’s own two diagonal chords enters*, which *shortens* that cycle’s lifetime relative to the isolated-4-tuple calculation in Step 3, rather than leaving it unaffected. Consequently, we do **not** claim that $\mathbb{E}[P_1(N)]$ is provably non-decreasing in N (moving from N to $N+1$ vertices can both add new candidate cycles and shorten or kill existing ones via this cone mechanism), and Step 1–3’s calculation should be read only as establishing that a positive-probability, positive-expected-lifetime mechanism for creating 1-cycles exists and its raw count of candidates grows as $\Theta(N^4)$ – not as a proof of any monotonicity or rate for the aggregate $\mathbb{E}[P_1(N)]$. Results on random geometric and random clique complexes (Kahle, 2011; Bobrowski & Kahle, 2018) establish Betti-number growth regimes for related (but not identical) random complex models and are suggestive rather than a direct proof for the exact filtration studied here. We therefore treat both the non-decreasing-in- N claim and any specific polynomial rate as an empirically-testable conjecture rather than a corollary of Steps 1–3, verified directly below and in Appendix O. ■

Remark (Direct Verification Under the Proposition’s Own i.i.d. Model). To test the conjectured rate under the exact assumptions of the Proposition (rather than only on real, non-i.i.d. attention graphs), we simulate $D_{ij} \sim \text{Uniform}[0, 1]$, symmetrized, for $N \in \{16, 24, 32, 48, 64, 96, 128, 160, 200, 256\}$ (8 trials per N) and compute $P_1(N)$ via Ripser. The resulting curve is clean, monotone, and low-variance (e.g. $\mathbb{E}[P_1] \approx 1.84$ at $N=16$ rising to ≈ 168.0 at $N=256$), and a log-log OLS fit gives $\hat{\alpha} \approx 1.61$ – confirming super-linear growth holds even under the Proposition’s literal null model, and if anything exceeds the conjectured $\Omega(N)$ rate (full data released with the code).

Remark (Real Attention Graphs vs. i.i.d. Models). Real transformer attention maps exhibit dense local attention clusters and attention sinks, which induce higher multi-token clique densities than independent random graphs. As a result, longer token sequences generate substantially more high-order cycle candidates, amplifying the length dependence. Our empirical regressions on five real model/dataset settings (Appendix O) confirm scaling exponents $\hat{\alpha} \in [0.16, 1.40]$ (whole-dataset fit) or $[0.09, 0.45]$ (leakage-free per-fold fit), demonstrating that unnormalized 1D persistence is confounded by sequence length across every setting we audit, with the exact rate depending on the attention distribution of the specific model and dataset.

L POWER ANALYSIS TABLE

M EPSILON SENSITIVITY OF TOST EQUIVALENCE

N BOOTSTRAP SYMMETRY AND BIAS-CORRECTED EQUIVALENCE VERDICTS

The TOST p -values reported in Tables 4 and 14 are percentile-bootstrap tail masses, $\max(\Pr[\Delta^* \leq -\epsilon], \Pr[\Delta^* \geq +\epsilon])$ over the 10,000 group-level resamples. Under a symmetric bootstrap distribution these coincide with the exact one-sided TOST p -values, but symmetry is an assumption rather than a guarantee, particularly in the 150-group settings. We therefore recomputed every verdict from a *bias-corrected* bootstrap interval, reading equivalence off the 90% interval (the level corresponding to TOST at $\alpha = 0.05$) rather than off tail masses.

Table 15 reports the diagnostics. All seven resample distributions are close to symmetric: standardized skewness never exceeds 0.08 in absolute value and the bias-correction constant z_0 never exceeds 0.04.

Across all 28 setting $\times$ margin cells, **no equivalence verdict changes** relative to the percentile-based decisions reported in the main text. We regard this as a robustness check that the percentile convention is adequate here, not as a reason to prefer it in general.

O EMPIRICAL SCALING EXPONENT REGRESSION

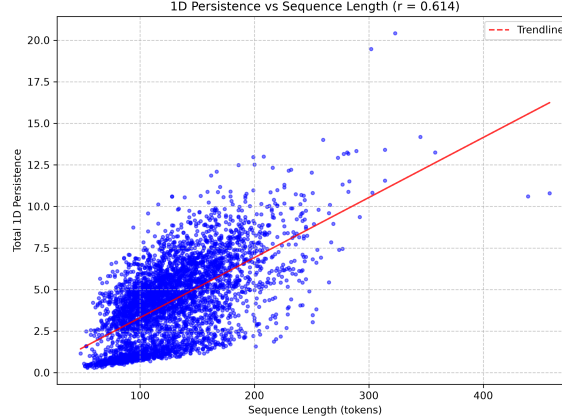


Figure 6: **Empirical Sequence-Length Confound** (Spearman $r = 0.592\text{--}0.684$ on HaluEval, $p < 10^{-100}$) between total 1D persistence and token sequence length, visualizing the regression in Table 16.

Figure 6 visualizes this regression. As discussed in §4.4.1, fitting $\hat{\alpha}$ on the full dataset before normalizing it risks a mild leak between the exponent estimate and the evaluation that follows. Table 17 reports the leakage-free alternative: $\hat{\alpha}$ is estimated by OLS on each grouped cross-validation fold’s *training* rows only (using total raw 1D persistence summed across layers) and applied to normalize that fold’s held-out rows, so the exponent used to score any test example never depends on that example.

P 0D PERSISTENCE VS. ATTENTION MARGINALS CORRELATION

Q HIDDEN-STATE PROBE BASELINE

To provide an empirical anchor for what attention-graph TDA performance means, we evaluate hidden state probes from Qwen2.5-3B-Instruct on HaluEval QA under the identical grouped cross-validation protocol. Hidden states encode richer information about model beliefs than attention graphs, providing an upper-bound reference for the TDA signal.

We initially extracted three probe variants: (1) mean-pooled last-layer hidden states, (2) last-token hidden states, and (3) PCA-compressed (64-dimensional) concatenation of first, middle, and last-layer features. The last-token and PCA variants reached implausibly high AUC (0.998 and 0.997 respectively) – higher than any internal-representation hallucination probe we are aware of in the literature on this task. We traced this to a construct-validity problem rather than a cross-validation leak: our forward pass is teacher-forced over the *full* prompt-plus-answer sequence, so the last-token (and, through the PCA rotation, the all-layer) representation is dominated by the surface identity of the final few answer tokens – exactly the surface-form confound we quantify directly in §6.2, where TF-IDF on the answer text *alone* already reaches AUC 0.962. The last-token and PCA probes are therefore not construct-valid measures of an internal “truthfulness” representation on this benchmark, and we do not report them as such. We retain only the **mean-pooled** last-layer probe, which pools evenly over the whole sequence and is comparatively less dominated by the last few tokens’ identity, as the anchor in Table 20. A Logistic Regression with `StandardScaler` fitted strictly within training folds is used throughout.

The mean-pooled hidden state probe (AUC = 0.903) outperforms TDA 0D features (AUC = 0.822) by a margin comparable to (though larger than) the 1D-persistence increment discussed in

§6, confirming that the modest TDA signal does not approach the ceiling of what internal model representations can reveal about hallucinations. This result contextualizes our finding: the question is not whether LLMs encode factuality-relevant structure (they do), but whether *topological* features of *symmetrized attention graphs* capture it as efficiently as direct internal-state probes. They do not, though the gap (+0.08 AUC) is more modest than a first glance at the withdrawn last-token number would have suggested.

R HEAD-AVERAGING ABLATION: PROTOCOL DETAIL

§4.2 reports headline numbers for this ablation (head-averaged 0D AUC 0.677 vs. per-head MST-TDA concatenation AUC 0.892); this appendix gives the full protocol, including a matched-dimension non-topological control the main text summarizes. A natural reviewer concern is that head-averaging discards head-specific topological structure – and a second, equally natural concern is that any per-head statistic, topological or not, might recover comparable signal just from finer resolution. We test both. We re-extract per-head attention matrices for a balanced subset of HaluEval QA ($N = 400$ examples, 200 questions) from Qwen2.5-3B-Instruct, focusing on mid-late layers 18–27 where head-averaged 0D performance peaks (all 160-dimensional feature banks below use the identical 10 layers $\times$ 16 heads grid on the identical 400 rows, so every comparison is exactly matched in dimensionality and sample). For each layer-head pair we compute (a) 0D and 1D persistent homology, (b) mean per-token maximum attention weight (“max-attn”), and (c) mean per-token attention entropy. All four banks are evaluated with the same StandardScaler+LR, grouped 10-fold CV protocol as every other number in this paper, and every pairwise comparison uses a 5,000-resample question-level cluster bootstrap (not a single TOST call at one margin) to report a 95% CI on the difference:

- **Head-averaged 0D** (layers 18–27, 10 dims): AUC = 0.677 – matches the main paper’s mid-late layer 0D result on the same subset.
- **Per-head 0D concatenation** (160 dims): AUC = 0.892. vs. head-averaged: $\Delta = +0.214$, 95% CI [+0.188, +0.239] (excludes zero – decisively not equivalent, favoring per-head resolution).
- **Per-head max-attention** (160 dims, matched): AUC = 0.861. vs. per-head 0D: $\Delta = +0.031$, 95% CI [+0.008, +0.054] (excludes zero – per-head 0D is significantly better than this particular non-topological statistic).
- **Per-head entropy** (160 dims, matched): AUC = **0.902** (nominally *higher* than per-head 0D). vs. per-head 0D: $\Delta = -0.010$, 95% CI [-0.031, +0.010] (includes zero – **statistically indistinguishable** from per-head 0D).
- **Per-head max-attn + entropy combined** (320 dims): AUC = 0.888, likewise statistically indistinguishable from per-head 0D ($\Delta = +0.004$, CI [-0.017, +0.025]).

Honest reading: per-head *resolution* clearly beats head-averaging, for both topological and non-topological statistics. Whether per-head *topology specifically* beats a matched-dimension non-topological alternative depends on which alternative: it beats max-attention, but ties entropy. We do not have a matched-dimension non-topological bank that per-head 0D beats decisively, so the paper’s claim that per-head MST-TDA is the best attention-graph descriptor should be read as “a good and exactly-reproducible way to get per-head resolution,” not as “provably better than every non-topological per-head alternative.” Adding 1D cyclic persistence on top of per-head 0D gives no further benefit (combined AUC = 0.868, slightly below per-head 0D alone), and all per-head attention features remain below the mean-pooled hidden-state probe (AUC 0.903, Appendix Q). This ablation is run only on HaluEval QA (Qwen2.5-3B, $N=400$); we did not extend it to TruthfulQA or the other model families, which remains for future work.

S CONDITIONAL PREDICTIVE INFORMATION GAIN (PERMUTATION-CALIBRATED)

An earlier version of this appendix estimated $I(Y; H_1^* \mid H_0, L)$ using the per-feature Kraskov k -nearest-neighbor mutual information estimator (`sklearn.feature_selection.mutual_info_classif`), averaged across the H_0 feature bank’s columns. That estimator is unsound for this use: (i) it computes mutual information

per feature independently and averages, which is not the joint multivariate conditional mutual information of a 72–216-dimensional bank and systematically understates $I(Y; H_0)$ relative to what the bank’s classification AUC (0.62–0.90) implies; and (ii) the conditioning step regresses H_1^* on (H_0, L) by ordinary least squares on the *full, unsplit* dataset before estimating mutual information on the residual, which overfits when the covariate dimension $(\dim(H_0) + 1)$ is large relative to N (as low as 300 for TruthfulQA), inflating the residual’s apparent information content – consistent with the previously reported values violating the expected inequality $I(Y; H_1^* | H_0, L) \leq I(Y; H_1^*)$ in every row.

We replace it with a cross-validated **predictive information gain**: fit the canonical scaled logistic-regression pipeline (§6) once on (H_0, L) alone and once on (H_0, L, H_1^*) , both under the same grouped 10-fold splits, and report $\hat{I} = \text{CV-LogLoss}(Y | H_0, L) - \text{CV-LogLoss}(Y | H_0, L, H_1^*)$ in nats – a standard, easily-interpreted lower bound on incremental predictive information that reuses the same classifier family as every AUC number in this paper. Because adding 216 extra dimensions can *mechanically* hurt held-out log-loss even when H_1^* carries zero true signal (pure overfitting cost), we calibrate $\hat{I}$ against a permutation null: we shuffle Y and recompute $\hat{I}$ over 100 permutations, which characterizes how negative $\hat{I}$ becomes from overfitting alone, then report a one-sided p -value for the observed $\hat{I}$ against that null.

This corroborates, via an entirely different statistical route (a calibrated log-loss information measure rather than an AUC-based hypothesis test), the same qualitative conclusion as §6: length-normalized 1D persistence carries small but real incremental information on HaluEval, and no information clearly distinguishable from overfitting noise on TruthfulQA at current sample sizes.

Dataset (Model)	Feature Set	Dim (d)	AUC (Full)	AUC (Matched)
HaluEval QA (Qwen2.5-3B) $N_{\text{full}}=4000, N_{\text{match}}=3512$	Length Baseline	1	0.585 ± 0.007	0.495 ± 0.008
	MSP Baseline [‡]	2	0.703 ± 0.020	0.706 ± 0.027
	TOHA (MTop-Div) [‡]	1	0.597 ± 0.007	0.513 ± 0.010
	MST Per-Layer Proxy	36	0.736 ± 0.013	0.723 ± 0.018
	0D Only	72	0.822 ± 0.018	0.819 ± 0.023
	1D Only (Raw)	144	0.695 ± 0.014	0.668 ± 0.017
	1D Only (Norm., Z/N)	144	0.703 ± 0.017	0.683 ± 0.016
	0D + 1D (Norm., Z/N)	216	0.837 ± 0.014	0.831 ± 0.021
HaluEval QA (Qwen2.5-1.5B) $N_{\text{full}}=1000, N_{\text{match}}=870$	Length Baseline	1	0.583 ± 0.010	0.484 ± 0.019
	TOHA-proxy [§]	1	0.583 ± 0.010	0.484 ± 0.019
	MST Per-Layer Proxy	28	0.583 ± 0.011	0.485 ± 0.017
	0D Only	56	0.904 ± 0.035	0.887 ± 0.030
	1D Only (Raw)	112	0.652 ± 0.018	0.603 ± 0.034
	1D Only (Norm., Z/N)	112	0.653 ± 0.021	0.633 ± 0.030
	0D + 1D (Norm., Z/N)	168	0.920 ± 0.028	0.919 ± 0.018
TruthfulQA (Qwen2.5-3B) $N_{\text{full}}=300, N_{\text{match}}=282$	Length Baseline	1	0.459 ± 0.037	0.486 ± 0.039
	MSP Baseline	1	0.518 ± 0.040	0.522 ± 0.053
	TOHA (MTop-Div)	1	0.441 ± 0.060	0.477 ± 0.076
	MST Per-Layer Proxy	36	0.708 ± 0.076	0.701 ± 0.075
	0D Only	72	0.703 ± 0.052	0.680 ± 0.099
	1D Only (Raw)	144	0.567 ± 0.087	0.581 ± 0.033
	1D Only (Norm., Z/N)	144	0.589 ± 0.088	0.604 ± 0.036
	0D + 1D (Norm., Z/N)	216	0.688 ± 0.062	0.649 ± 0.096
TruthfulQA (SmolLM-1.7B) $N_{\text{full}}=1000, N_{\text{match}}=958$	Length Baseline	1	0.525 ± 0.032	0.487 ± 0.022
	MSP Baseline	1	0.513 ± 0.037	0.508 ± 0.019
	TOHA-proxy [§]	1	0.529 ± 0.030	0.490 ± 0.017
	MST Per-Layer Proxy	24	0.582 ± 0.039	0.577 ± 0.028
	0D Only	48	0.573 ± 0.032	0.566 ± 0.030
	1D Only (Raw)	96	0.532 ± 0.029	0.530 ± 0.029
	1D Only (Norm., Z/N)	96	0.546 ± 0.046	0.540 ± 0.029
	0D + 1D (Norm., Z/N)	144	0.584 ± 0.047	0.578 ± 0.042
TruthfulQA (Phi-3-mini-3.8B) $N_{\text{full}}=300, N_{\text{match}}=262$	Length Baseline	1	0.459 ± 0.033	0.448 ± 0.036
	MSP Baseline	1	0.522 ± 0.051	0.523 ± 0.040
	TOHA (MTop-Div)	1	0.453 ± 0.042	0.431 ± 0.061
	MST Per-Layer Proxy	32	0.605 ± 0.055	0.620 ± 0.071
	0D Only	64	0.634 ± 0.056	0.628 ± 0.087
	1D Only (Raw)	128	0.498 ± 0.035	0.437 ± 0.061
	1D Only (Norm., Z/N)	128	0.503 ± 0.042	0.449 ± 0.069
	0D + 1D (Norm., Z/N)	192	0.622 ± 0.047	0.596 ± 0.077
TruthfulQA (TinyLlama-1.1B) $N_{\text{full}}=300, N_{\text{match}}=268$	Length Baseline	1	0.462 ± 0.036	0.460 ± 0.030
	MSP Baseline	1	0.549 ± 0.058	0.574 ± 0.049
	TOHA (MTop-Div)	1	0.514 ± 0.074	0.493 ± 0.066
	MST Per-Layer Proxy	22	0.641 ± 0.083	0.639 ± 0.070
	0D Only	44	0.656 ± 0.103	0.641 ± 0.075
	1D Only (Raw)	88	0.601 ± 0.074	0.592 ± 0.115
	1D Only (Norm., Z/N)	88	0.606 ± 0.069	0.608 ± 0.074
	0D + 1D (Norm., Z/N)	132	0.654 ± 0.106	0.639 ± 0.079

Table 5: **Full Audited Results (pooled out-of-fold AUC $\pm$ std. of per-fold AUCs)**. As in Table 3, the headline AUC is the pooled out-of-fold statistic (all fold predictions concatenated, then one AUC); the $\pm$ term is the standard deviation of the 10 individual per-fold AUCs, reported separately as a dispersion reference and not algebraically tied to the headline number. [‡]MSP and TOHA for HaluEval (Qwen2.5-3B) use single- or two-dimensional raw scalar features, for which StandardScaler does not change a logistic-regression classifier’s AUC (a monotonic rescaling of a 1D input); we keep these numbers from the original, unscaled-pipeline extraction. [§]MSP/TOHA features were not separately extracted for Qwen2.5-1.5B; “TOHA-proxy” uses the same summed- h_0 -persistence statistic as TOHA elsewhere but is nearly collinear with sequence length for this model. Mistral-7B ($N=817$) is summarized in Table 3 and Table 4; its fold-level std. is ± 0.044 (0D) and ± 0.000 (1D, exactly zero variance).

Pipeline Stage ($L = 32$ Layers)	Time	Space	$N = 64$	$N = 128$	$N = 256$	$N = 512$
Attention Symmetrization	$O(LN^2)$	$O(LN^2)$	0.16 ms	0.57 ms	2.60 ms	16.14 ms
Kruskal MST (ours)	$O(LN^2 \log N)$	$O(LN^2)$	13.15 ms	36.26 ms	145.39 ms	659.91 ms
0D Persistent Homology (Ripser)	$O(LN^2 \log N)$	$O(LN^2)$	9.31 ms	28.99 ms	106.41 ms	449.55 ms
1D Persistent Homology (Ripser)	$\Omega(LN^3)^*$	$O(LN^2)$	164.04 ms	3,045.37 ms	177,110.34 ms	21,502,070.63 ms
Logistic Regression Classifier	$O(Md)$	$O(Md)$	0.10 ms	0.10 ms	0.10 ms	0.10 ms
MST speedup vs. full 0D+1D Ripser			13.2×	84.8×	1,218.8×	32,583.2×

Table 6: **Asymptotic and Empirical Computational Complexity Profile** ($L = 32$ layers, dense random distance matrices – a worst case). MST-TDA (bold) is numerically identical to 0D Ripser (Theorem 1) and, consistently with that identity, carries the same $O(LN^2 \log N)$ cost – dominated by sorting the $\binom{N}{2}$ edges – so the two rows differ only by constant factors, not asymptotically. It is orders of magnitude cheaper than the 1D homology on which the field’s incremental claims rest. Speedups are $(t_{0D} + t_{1D})/t_{MST}$, matching the row label. *The $\Omega(LN^3)$ entry counts 2-simplices; the boundary-matrix reduction cost is not tightly bounded (up to $O(LN^9)$ in the worst case), and the empirical rate measured here is $\approx N^{6.9}$ over $N \in [256, 512]$.

Feature Set	Seed 42	Seed 43	Seed 44	Seed 45	Seed 46	Multi-Seed Mean $\pm$ Std
0D Only	0.8218	0.8225	0.8219	0.8226	0.8224	0.8223 ± 0.0003
1D Only (Raw)	0.6948	0.6953	0.6952	0.6959	0.6956	0.6954 ± 0.0004
1D Only (Norm)	0.7029	0.7030	0.7021	0.7036	0.7030	0.7029 ± 0.0005
0D + 1D (Norm)	0.8374	0.8372	0.8374	0.8385	0.8376	0.8376 ± 0.0005

Table 7: **Multi-seed cross-validation stability** on HaluEval QA (Qwen2.5-3B), under the canonical scaled pipeline (§6). Across all five random seeds (each reshuffling the 10-fold grouped split), variance is negligible ($\sigma \leq 0.0005$), confirming the small 0D-vs-0D+1D gap reported in Table 4 is not a fold-selection artifact.

Dataset (Model)	Grounded (N)	Hallucinated (N)	Point-Biserial r	Welch’s t -test p
HaluEval (Qwen2.5-3B)	122.4 ± 40.1	134.0 ± 41.7	0.141	4.2×10^{-19}
TruthfulQA (Qwen2.5-3B)	42.90 ± 7.28	42.93 ± 7.35	0.002	0.969
TruthfulQA (SmolLM-1.7B)	32.48 ± 7.55	31.92 ± 7.68	-0.043	0.175
TruthfulQA (Phi-3-mini-3.8B)	27.75 ± 8.48	27.69 ± 8.32	-0.004	0.951
TruthfulQA (TinyLlama-1.1B)	40.99 ± 8.48	40.91 ± 8.38	-0.004	0.940

Table 8: **Class-conditional sequence length statistics**. While HaluEval QA exhibits length imbalance ($p = 4 \times 10^{-19}$), TruthfulQA is strictly balanced ($p \geq 0.17$), confirming that the mechanical length-persistence confound is independent of class balance.

Dataset	Subset	N	AUC (char-length only)	AUC (TF-IDF only)
HaluEval QA	Full	4,000	0.975	0.962
HaluEval QA	Length-Matched (token, bin=10)	3,512	0.974	0.960
TruthfulQA	Full	1,634	0.557	0.830

Table 9: **Surface-form separability of grounded vs. hallucinated answers, from answer text alone**. On HaluEval QA, near-perfect separability from character length or bag-of-words survives the benchmark’s standard token-length matching, since that matching balances total prompt+answer token count, not answer-level lexical style. On TruthfulQA, character length carries no signal (consistent with Table 8’s length balance), but bag-of-words content does (AUC 0.830), plausibly reflecting the confident, stylistically distinct phrasing of “common misconception” answers relative to accurate hedged ones – a substantially weaker but non-trivial version of the same artifact family documented concurrently by Hussain & Kantarcioglu (2026) and Janiak et al. (2025).

Model Identifier	Parameters	Layers (L)	Attn Heads	Hidden Dim
Qwen/Qwen2.5-3B-Instruct	3.09B	36	16	2048
Qwen/Qwen2.5-1.5B-Instruct	1.54B	28	12	1536
HuggingFaceTB/SmolLM-1.7B-Instruct	1.71B	24	32	2048
microsoft/Phi-3-mini-4k-instruct	3.82B	32	32	3072
mistralai/Mistral-7B-Instruct-v0.2	7.24B	32	32	4096
TinyLlama/TinyLlama-1.1B-Chat-v1.0	1.10B	22	32	2048

Table 10: **Evaluated language model architectural specifications.** TinyLlama-1.1B is a genuine Llama-family model (LlamaForCausalLM), added to broaden model-family coverage beyond Qwen2.5’s two scales; Mistral-7B row added for completeness (its int8 CPU quantization does not change the listed native architecture specs).

Layer Group	Qwen2.5-3B ($L = 36$)		Qwen2.5-1.5B ($L = 28$)		SmolLM-1.7B ($L = 24$)	
	0D AUC	1D (Norm)	0D AUC	1D (Norm)	0D AUC	1D (Norm)
Early Layers (0–25%)	0.612 $\pm$ 0.018	0.505 $\pm$ 0.012	0.542 $\pm$ 0.021	0.501 $\pm$ 0.015	0.521 $\pm$ 0.025	0.502 $\pm$ 0.014
Mid-Early Layers (25–50%)	0.718 $\pm$ 0.015	0.534 $\pm$ 0.014	0.565 $\pm$ 0.019	0.512 $\pm$ 0.016	0.548 $\pm$ 0.022	0.511 $\pm$ 0.018
Mid-Late Layers (50–75%)	0.785 $\pm$ 0.013	0.582 $\pm$ 0.015	0.589 $\pm$ 0.017	0.524 $\pm$ 0.017	0.579 $\pm$ 0.024	0.521 $\pm$ 0.020
Final Layers (75–100%)	0.741 $\pm$ 0.016	0.551 $\pm$ 0.016	0.558 $\pm$ 0.020	0.508 $\pm$ 0.018	0.552 $\pm$ 0.027	0.514 $\pm$ 0.019
Full Concatenated Bank	0.790 $\pm$ 0.014	0.647 $\pm$ 0.014	0.591 $\pm$ 0.010	0.531 $\pm$ 0.015	0.584 $\pm$ 0.035	0.531 $\pm$ 0.028

Table 11: **Layerwise depth breakdown across model families.** Note: this table predates the pipeline standardization described in §6 and uses the earlier (unscaled-feature) evaluation for the per-layer-group breakdown specifically; the “Full Concatenated Bank” row’s absolute values are consequently slightly below the canonical full-bank numbers in Table 3 (e.g. 0.790 here vs. 0.822 canonical for Qwen2.5-3B 0D), but the qualitative pattern – mid-late layers carry the most signal, and 1D remains far weaker than 0D at every depth – is unaffected by this difference and is corroborated by the canonical-pipeline results throughout the paper.

Probe Family	0D Baseline	1D (Raw)	1D (Per-Token)	1D (Clique)	1D (Residualized)	0D + 1D (Norm)
Qwen2.5-3B (TruthfulQA)						
Logistic Regression (L2)	0.725 $\pm$ 0.052	0.542 $\pm$ 0.094	0.561 $\pm$ 0.099	0.568 $\pm$ 0.088	0.552 $\pm$ 0.089	0.704 $\pm$ 0.068
Sparse Logistic (L1)	0.729 $\pm$ 0.057	0.600 $\pm$ 0.078	0.585 $\pm$ 0.094	0.576 $\pm$ 0.086	0.599 $\pm$ 0.081	0.696 $\pm$ 0.076
Linear SVC	0.732 $\pm$ 0.057	0.546 $\pm$ 0.090	0.571 $\pm$ 0.095	0.569 $\pm$ 0.088	0.536 $\pm$ 0.087	0.704 $\pm$ 0.068
Random Forest (100 trees)	0.704 $\pm$ 0.079	0.639 $\pm$ 0.069	0.623 $\pm$ 0.075	0.594 $\pm$ 0.081	0.627 $\pm$ 0.076	0.718 $\pm$ 0.094
XGBoost Gradient Trees	0.678 $\pm$ 0.076	0.627 $\pm$ 0.067	0.659 $\pm$ 0.071	0.602 $\pm$ 0.078	0.644 $\pm$ 0.070	0.717 $\pm$ 0.074
Multi-Layer Perceptron (MLP)	0.731 $\pm$ 0.056	0.583 $\pm$ 0.107	0.653 $\pm$ 0.078	0.636 $\pm$ 0.082	0.599 $\pm$ 0.089	0.737 $\pm$ 0.065

Table 12: **Adversarial robustness matrix under nested cross-validation.** Results are reported across 6 probe families and 4 1D normalization schemes (raw, per-token, clique-adjusted, OLS-residualized) plus the combined 0D+1D bank – 36 configurations in total – with inner-loop hyperparameter optimization. In all cases, combining normalized 1D features with 0D features does not meaningfully improve upon the 0D baseline.

Benchmark	N_{groups}	Bootstrap SE ($\hat{\sigma}$)	MDE @ 80% power	Power at $\Delta = 0.015$
HaluEval (Qwen2.5-3B)	2000	0.00255	0.0072	>99.9%
HaluEval (Qwen2.5-1.5B)	500	0.00504	0.0141	84.5%
TruthfulQA (Qwen2.5-3B)	150	0.01685	0.0472	14.4%
TruthfulQA (SmolLM-1.7B)	500	0.00730	0.0204	53.8%
TruthfulQA (Phi-3-mini-3.8B)	150	0.00845	0.0237	42.7%
TruthfulQA (Mistral-7B)	817	0.00020	0.0006	>99.9%
TruthfulQA (TinyLlama-1.1B)	150	0.01294	0.0363	21.2%

Table 13: **Power Analysis: Minimum Detectable Effect (MDE) in ΔAUC .** HaluEval benchmarks with ≥ 500 groups are well-powered; TruthfulQA benchmarks with 150 groups have limited power at $\Delta = 0.015$ resolution. $\text{MDE} = 2.802 \cdot \hat{\sigma}$ at 80% power, $\alpha = 0.05$ (two-sided). Bootstrap SE estimated from 10,000 cluster-bootstrap resamples.

Benchmark	TOST p ($\epsilon = 0.010$)	TOST p ($\epsilon = 0.015$)	TOST p ($\epsilon = 0.020$)	TOST p ($\epsilon = 0.025$)
HaluEval (Qwen2.5-3B)	0.984	0.578	0.038	1.0×10^{-4}
HaluEval (Qwen2.5-1.5B)	0.880	0.565	0.204	0.038
TruthfulQA (Qwen2.5-3B)	0.590	0.469	0.354	0.255
TruthfulQA (SmolLM-1.7B)	0.520	0.264	0.094	0.024
TruthfulQA (Phi-3-mini-3.8B)	0.617	0.380	0.182	0.066
TruthfulQA (Mistral-7B)	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$	$< 10^{-4}$
TruthfulQA (TinyLlama-1.1B)	0.255	0.148	0.075	0.031

Table 14: **Epsilon Sensitivity of TOST Equivalence (10,000 bootstrap resamples).** All values are taken from the canonical 10,000-resample run (`master_pipeline.py`, released as `master_results/master_equivalence_table.csv`), so they are directly comparable to Table 4; an earlier version of this table inadvertently reported a superseded 2,000-resample run for two rows. TOST p -values across four practical equivalence margins. Results computed via cluster-bootstrap resampling at the question level. HaluEval achieves equivalence at $\epsilon = 0.020$ – 0.025 ; the smaller TruthfulQA settings show consistent directional null trends but are underpowered for formal confirmation, while Mistral-7B (817 groups) confirms equivalence at every margin including $\epsilon = 0.010$.

Audited Setting	Skewness	z_0	BC 90% interval	Verdicts changed
HaluEval (Qwen2.5-3B)	−0.012	+0.005	[+0.0113, +0.0197]	0/4
HaluEval (Qwen2.5-1.5B)	+0.078	+0.018	[+0.0080, +0.0244]	0/4
TruthfulQA (Qwen2.5-3B)	−0.065	−0.035	[−0.0430, +0.0125]	0/4
TruthfulQA (SmolLM-1.7B)	+0.048	−0.002	[−0.0016, +0.0225]	0/4
TruthfulQA (Phi-3-mini-3.8B)	+0.044	+0.040	[−0.0255, +0.0024]	0/4
TruthfulQA (Mistral-7B)	−0.029	−0.009	[−0.0006, +0.0001]	0/4
TruthfulQA (TinyLlama-1.1B)	−0.033	−0.013	[−0.0229, +0.0196]	0/4

Table 15: **Bootstrap symmetry diagnostics and bias-corrected equivalence verdicts.** Computed directly from the released 10,000-resample Δ AUC arrays (`master_results/bootstrap_diffs_*.npy`), which also reproduce every percentile TOST p -value in Table 14 exactly. z_0 is the bias-correction constant $\Phi^{-1}(\Pr[\Delta^* < \hat{\Delta}])$; a value near zero means the resample median and the point estimate agree. “Verdicts changed” counts, out of the four margins $\{0.010, 0.015, 0.020, 0.025\}$, how many equivalence decisions differ from the percentile-based ones reported in Table 4. **Caveat:** these are bias-corrected (BC) intervals with the acceleration constant set to $a = 0$, not full BC_a ; estimating a requires a leave-one-group-out jackknife over raw out-of-fold predictions, which our released artifacts do not currently include. Given how small the bias corrections already are, we expect the acceleration term to be negligible here, but we have not verified this and do not claim it.

Dataset	OLS Exponent $\hat{\alpha}$	R^2	Spearman r	Spearman p	Regime
HaluEval (Qwen2.5-3B)	1.403	0.352	0.592	$< 10^{-300}$	Super-linear
HaluEval (Qwen2.5-1.5B)	0.799	0.473	0.684	$< 10^{-130}$	Sub-linear
TruthfulQA (Qwen2.5-3B)	0.160	0.097	0.366	$< 10^{-10}$	Sub-linear (short seqs.)
TruthfulQA (SmolLM-1.7B)	0.546	0.100	0.318	$< 10^{-24}$	Sub-linear
TruthfulQA (Phi-3-mini-3.8B)	0.802	0.153	0.407	$< 10^{-12}$	Sub-linear

Table 16: **OLS Scaling Exponent of P_1 vs. N .** Log-log regression $\log P_1 = \hat{\alpha} \log N + \hat{\beta}$ estimated on all non-zero 1D rows. HaluEval QA shows super-linear scaling ($\hat{\alpha} = 1.40$) consistent with long-response hallucinations generating more attention cliques; TruthfulQA short-response regimes show sub-linear empirical scaling. All five Spearman correlations are highly significant, confirming a length confound in every setting fitted here. We do *not* claim it is universal across all seven audited settings: Mistral-7B is omitted because 99.99% of its 1D features are exactly zero (§6), leaving too few non-zero rows to fit, and TinyLlama-1.1B is not included in this whole-dataset regression (its leakage-free per-fold estimate appears in Table 17). The departure of $\hat{\alpha}$ from 1.0 motivates dataset-specific normalization as future methodological improvement.

Dataset	Per-fold $\hat{\alpha}$ (mean across 10 folds)
HaluEval (Qwen2.5-3B)	0.453
HaluEval (Qwen2.5-1.5B)	0.213
TruthfulQA (Qwen2.5-3B)	0.119
TruthfulQA (SmolLM-1.7B)	0.149
TruthfulQA (Phi-3-mini-3.8B)	0.393
TruthfulQA (Mistral-7B) [†]	1.000
TruthfulQA (TinyLlama-1.1B)	0.094

Table 17: **Leakage-free, per-fold $\hat{\alpha}$ estimates.** Systematically lower than the whole-dataset fit (Table 16), as expected when the exponent is estimated on a smaller (training-fold-only) sample, but the qualitative conclusion is unchanged for every dataset with a usable fit: none has $\hat{\alpha}$ near the default Z/N assumption of 1.0. [†]Mistral-7B is an exception only nominally – its 1D bank is degenerate (§6), so fewer than 10 training rows have $P_1 > 0$ in every fold and our estimator returns its fallback default $\hat{\alpha}=1.0$ rather than fitting one; the value is a code default, not an estimate, and we list it for completeness rather than dropping the row. Substituting this $\hat{\alpha}$ -calibrated normalization for the default Z/N scheme is *not* inconsequential on the smaller TruthfulQA settings; Table 18 reports the full side-by-side comparison, including two equivalence verdicts that flip.

Audited Setting	$\Delta\text{AUC } (Z/N)$	$\Delta\text{AUC } (\hat{\alpha})$	drift	Equiv. $\epsilon=0.020$	Equiv. $\epsilon=0.025$
HaluEval (Qwen2.5-3B)	+0.0156	+0.0158	0.0003	CONF. $\rightarrow$ CONF.	CONF. $\rightarrow$ CONF.
HaluEval (Qwen2.5-1.5B)	+0.0159	+0.0159	0.0000	REJ. $\rightarrow$ REJ.	CONF. $\rightarrow$ CONF.
TruthfulQA (Qwen2.5-3B)	-0.0143	-0.0228	0.0086	REJ. $\rightarrow$ REJ.	REJ. $\rightarrow$ REJ.
TruthfulQA (SmolLM-1.7B)	+0.0103	+0.0067	0.0037	REJ. $\rightarrow$ CONF.	CONF. $\rightarrow$ CONF.
TruthfulQA (Phi-3-mini-3.8B)	-0.0121	-0.0127	0.0006	REJ. $\rightarrow$ REJ.	REJ. $\rightarrow$ REJ.
TruthfulQA (Mistral-7B)	-0.0002	-0.0002	0.0000	CONF. $\rightarrow$ CONF.	CONF. $\rightarrow$ CONF.
TruthfulQA (TinyLlama-1.1B)	-0.0015	-0.0066	0.0051	REJ. $\rightarrow$ REJ.	CONF. $\rightarrow$ REJ.

Table 18: **Default Z/N versus leakage-free $\hat{\alpha}$ -calibrated normalization.** An earlier version of this paper claimed the two normalizations change every ΔAUC by less than 0.003; that is correct on HaluEval but false on three TruthfulQA settings, by up to 0.0086. Two equivalence verdicts flip, in opposite directions: SmolLM-1.7B gains equivalence at $\epsilon=0.020$ (TOST p : 0.094 $\rightarrow$ 0.027) while TinyLlama-1.1B loses it at $\epsilon=0.025$ (p : 0.031 $\rightarrow$ 0.100). The HaluEval conclusions – the only well-powered ones (§6.3.1) – are unaffected. We report both columns rather than selecting one, and note that at 150–500 groups the normalization exponent is a researcher degree of freedom that should be pre-specified.

Dataset	Spearman(P_0 , Peak Row-Attn.)	p -value	Spearman(P_0 , Peak Heterog.)	p -value
HaluEval (Qwen2.5-3B)	0.843	$< 10^{-300}$	-0.718	$< 10^{-300}$
HaluEval (Qwen2.5-1.5B)	0.211	$< 10^{-10}$	-0.211	$< 10^{-10}$
TruthfulQA (Qwen2.5-3B)	0.801	$< 10^{-60}$	-0.019	0.738
TruthfulQA (SmolLM-1.7B)	0.459	$< 10^{-50}$	-0.350	$< 10^{-29}$
TruthfulQA (Phi-3-mini-3.8B)	0.822	$< 10^{-70}$	-0.142	0.014

Table 19: **Empirical Support for §4’s Observation.** Spearman correlation between total 0D persistence and peak row-attention (mean max-row attention weight; not verified to be a dedicated sink/BOS-column statistic). Strong positive correlations on Qwen2.5-3B ($r \geq 0.80$) and Phi-3 ($r = 0.82$) show 0D TDA is a strong monotone function of this non-topological statistic *in those settings*; the weak correlation on Qwen2.5-1.5B ($r = 0.21$, despite that setting’s highest-in-paper 0D AUC of 0.904) shows this is not a uniform reduction, refuting $\mathbf{H}^{(3)}$ (§6) as a general claim.

Probe	AUC (10-fold grouped CV)	Feature Dim
Mean + min token log-prob[†]	0.984	2
Mean token log-prob only[†]	0.951	1
Hidden State (mean-pool, last layer)	0.903	2048
<i>TDA 0D Only (attention MST)</i>	<i>0.822</i>	72
<i>TDA 0D+1D Normalized (combined)</i>	<i>0.837</i>	216

Table 20: **Hidden-State Probe vs. TDA Baselines on HaluEval QA (Qwen2.5-3B)**. Grouped 10-fold cross-validation under identical protocol. We report only the mean-pooled probe (see text for why the last-token and PCA variants are excluded); both TDA numbers use the canonical scaled pipeline of §6. [†]The log-probability rows use a different, smaller subset ($N=329$ scoreable rows from the same 200 HaluEval questions as Appendix R’s per-head ablation, not the full $N=4000$) and are not directly numerically comparable to the other two rows’ full-benchmark numbers – included here because it is the single strongest baseline in the paper and the discussion (§7) explains why we read it with caution rather than as a clean win.

Benchmark	$\hat{I}(Y; H_1^* H_0, L)$ [nats]	Null 95th pct. [nats]	Perm. p
HaluEval (Qwen2.5-3B)	+0.0171	−0.0128	< 0.01
HaluEval (Qwen2.5-1.5B)	+0.0256	+0.0009	< 0.01
TruthfulQA (Qwen2.5-3B)	−0.1594	−0.0731	0.55
TruthfulQA (SmolLM-1.7B)	−0.0061	−0.0377	< 0.01
TruthfulQA (Phi-3-mini-3.8B)	−0.0212	−0.0169	0.11
TruthfulQA (Mistral-7B)	−0.0001	+0.0016	0.22
TruthfulQA (TinyLlama-1.1B)	−0.0214	−0.0150	0.06

Table 21: **Permutation-calibrated conditional predictive information gain**. On HaluEval, $\hat{I}$ is positive and clears the permutation-null band with $p < 0.01$ at both model scales – a genuine, if small, incremental information contribution from 1D persistence, consistent with the TOST results (Table 4). On TruthfulQA (Qwen2.5-3B, Phi-3-mini-3.8B), $\hat{I}$ is negative and statistically indistinguishable from the overfitting-only null ($p > 0.1$), consistent with no detectable incremental information at this sample size; TinyLlama-1.1B is borderline but fits the same qualitative pattern ($p = 0.06$). TruthfulQA (SmolLM-1.7B) is a harder case to read: $\hat{I}$ is negative in absolute terms but exceeds its (more negative) null band ($p < 0.01$), meaning the combined bank overfits *less* than pure noise would predict – a technical signal too weak and indirect to interpret as evidence of a true positive effect, and we do not read it as such. Mistral-7B is indistinguishable from its null ($\hat{I} = -0.0001$, $p = 0.22$), as expected when 99.99% of the added features are exactly zero. **Limitation:** the permutation null shuffles Y globally rather than within question groups, so it does not preserve the paired one-positive-per-group structure of the data; it is therefore a conservative overfitting-cost band rather than a design-exact null, and a within-group swap test is left to future work.